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Review article

Control theory in the era of intelligent systems: Applications, trends, and future directions

Maria Trigka *, Elias Dritsas

Department of Informatics and Computer Engineering, University of West Attica, Egaleo Park Campus, Athens, 12243, Greece

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ABSTRACT

Control theory has established itself as a fundamental discipline for the analysis and design of dynamical systems. From its classical foundations, including proportional–integral–derivative (PID) control, state–space representations, and stability analysis, it has progressively expanded toward advanced, robust, and predictive control frameworks. The increasing complexity of modern systems, characterized by large-scale integration, data-driven operations, and stringent safety requirements, is reshaping their methodological foundations. However, despite extensive progress, the field of control remains fragmented, and an integrative viewpoint connecting the different topics is still lacking. This survey provides a unified cross-domain perspective that consolidates established principles and emerging intelligent paradigms. Specifically, it addresses this research gap by synthesizing classical control theory with contemporary artificial intelligence (AI)-driven and cyber–physical systems (CPS) methodologies within a mathematically consistent framework. To the best of our knowledge, no prior survey has provided a unified analytical framework that jointly treats all the aforementioned topics. A distinctive contribution is the mathematically rigorous treatment of nonlinear observer design through Laguerre polynomial approximations, positioned in direct comparison with extended Kalman filters (EKF), high-gain observers (HGO), and hybrid estimation methods. Beyond the classical scope, this survey addresses the convergence of control with AI, machine learning (ML), deep learning (DL), Internet of Things (IoT) infrastructures, digital twins, and quantum-edge computing, emphasizing their implications for scalability, adaptability, and resilience. The novelty of this review lies in articulating an integrative framework that bridges robust analytical tools with intelligent, data-driven architectures, highlighting both methodological coherence and cross-domain applicability. Key challenges include managing nonlinear complexity, ensuring robustness under uncertainty, embedding ethical and governance-aware mechanisms, and bridging the gap between theoretical innovation and practical deployment. Future trends indicate a shift towards reinforcement learning (RL)-augmented control, hybrid physics–AI architectures, distributed CPS architectures, and sustainability-driven designs. By combining historical depth with forward-looking integration, this survey serves as both a consolidated reference and a roadmap for the next generation of intelligent and resilient control systems.

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* Corresponding author.
E-mail addresses: mtrigka@uniwa.gr (M. Trigka), idritsas@uniwa.gr (E. Dritsas).

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1. Introduction

Control theory forms the backbone of the analysis and regulation of dynamic systems. From its origins in feedback stabilization and PID controllers, this field has evolved into a unifying discipline spanning robust, adaptive, and distributed formulations. Classical concepts such as stability and observability remain fundamental; however, their application has shifted from local, low-order systems to large-scale infrastructures such as autonomous mobility, smart energy grids, and CPS (Glad & Ljung, 2018).

In parallel, system modeling has expanded beyond ordinary differential equations (ODE) and state-space formulations to include descriptor systems and differential–algebraic equations (DAE), which capture algebraic constraints in power, mechanical, and economic systems (Nise, 2020). These representations are now tightly coupled with estimation and observer design, extending the classical Kalman and Luenberger observers to generalized frameworks (Kumar & Gupta, 2023; Ríos-Ruiz et al., 2023). Simultaneously, the emergence of data-driven modeling and ML has introduced alternative strategies that enable control in domains where physical modeling is incomplete or intractable (Gong et al., 2023; Tang & Daoutidis, 2022).

The convergence of control with AI and ML is reshaping this field. RL, learning-augmented observers, and neural approximations increasingly complement traditional controllers, offering scalability and adaptability in uncertain networked environments (Recht, 2019; Wu et al., 2024). In this context, advanced observer design plays a central role: polynomial expansions, such as Laguerre functions, provide compact approximations that bridge operator-theoretic methods with learning-based estimators, situating nonlinear observer theory within the AI-control interface (Xu et al., 2024). This synergy positions control not only as a mature engineering discipline but also as an evolving framework for data-driven, resilient, and ethically governed intelligent systems (Yazdanpanah et al., 2023). For clarity and ease of reference, Table 1 provides a glossary of acronyms and abbreviations used in this study.

1.1. Motivation

This survey was motivated by the widening gap between classical control frameworks and the requirements of emerging intelligent and networked systems. While prior studies have primarily focused on PID and its extensions, robust control, or domain-specific implementations, few have systematically examined how AI and ML have reshaped the core notions of modeling, estimation, and observer design. This omission is critical because the integration of control with data-driven methodologies is becoming a defining requirement rather than a peripheral trend.

A distinctive aspect of this study is its emphasis on nonlinear observer formulations within this broader landscape, clarifying how classical estimation tools can be reinterpreted under a unified analytical perspective that aligns with modern learning-based control. Rather than proposing new estimation algorithms, this survey revisits existing observer structures through this integrative lens. For instance, Laguerre polynomial expansions are presented as a representative mathematical formalism within this landscape, illustrating how analytical tractability can coexist with data-informed system representations.

Overall, the motivation is to bridge the conceptual fragmentation between classical control theory topics and emerging AI-oriented methodologies, establishing an intellectual foundation upon which the subsequent contributions of this work are built.

Table 1
Glossary of acronyms and abbreviations.

Acronym	Definition
AI	Artificial Intelligence
ML	Machine Learning
RL	Reinforcement Learning
DL	Deep Learning
FL	Federated Learning
PI(D)	Proportional–Integral (Derivative)
EKF	Extended Kalman Filters
ARX	AutoRegressive with eXogenous input
REKF	Robust Extended Kalman Filter
RTAI	Real-Time Application Interface
HGO	High-Gain Observer
EHGO	Extended High-Gain Observer
(E)ALPIO	(Extended) ARX–Laguerre Proportional–Integral Observer
NARMAX	Nonlinear AutoRegressive Moving Average with eXogenous Inputs
NN	Neural Network
WRNN	Wavelet Recurrent NN
CReNN	Chebyshev Recurrent NN
DFNN	Dynamic Feedforward NN
MNN	Multilayer NN
ELM	Extreme Learning Machine
DC	Direct Current
MPC	Model Predictive Control
NMPC	Nonlinear MPC
MSE	Mean Squared Error
SBW	Steer-by-Wire
VCU	Vehicle Control Unit
CAN	Controller Area Network
LTI	Linear Time-Invariant
PDE	Partial Differential Equation
ODE	Ordinary Differential Equation
(A)ESR	(Adjusted) Error-to-Signal Ratio
CPS	Cyber–Physical Systems
IoT	Internet of Things
DAE	Differential–Algebraic Equations
PRISMA	Systematic Reviews and Meta-Analyses
MIMO	Multi-Input Multi-Output
FOPID	Fractional-Order PID
VOPID	Variable-Order PID
POD	Proper Orthogonal Decomposition
LQR	Linear Quadratic Regulators
LQG	Linear Quadratic Gaussian
PLC	Programmable Logic Controller
IAE	Integral Absolute Error
ISE	Integral Squared Error
ITSE	Integral of Time-weighted Squared Error
ITAE	Integral Time-weighted Absolute Error
PSO	Particle Swarm Optimization
FOPTD	First-Order-Plus-Dead-Time
SISO	Single Input, Single Output
UAV	Unmanned Aerial Vehicles
HIL	Hardware-in-the-Loop
VQA	Variational Quantum Algorithms
QAOA	Quantum Approximate Optimization Algorithm
EU	European Union
ISO	International Organization for Standardization
IEC	International Electrotechnical Commission
FDA	Food and Drug Administration
SaMD	Software as a Medical Device
QECC	Quantum Error-Correcting Code

1.2. Methodology

This survey adopts a structured, transparent, and reproducible methodology grounded in established evidence synthesis principles and aligned with the Systematic Reviews and Meta-Analyses (PRISMA) framework (Parums, 2021). The objective is to ensure systematic coverage of system modeling, observer design, and control architectures, while capturing their convergence with emerging technological paradigms such as AI-augmented estimation, cyber–physical infrastructures, digital-twin ecosystems, and quantum and edge-based control. No temporal restrictions were imposed, allowing the inclusion of both foundational and contemporary contributions that collectively illuminate the historical trajectory and current evolution of intelligent control systems.

A comprehensive literature search was conducted across IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and Google Scholar databases. Boolean expressions combined modeling descriptors including “control theory”, “state-space”, “transfer function”, “descriptor systems”, “differential–algebraic equations”, “nonlinear systems”, and “Laguerre functions”; observer design terms such as “Kalman filter”, “extended Kalman filter”, “high-gain observer”, “nonlinear observer”, “Laguerre observer”, “state estimation”, and “hybrid observers”; and controller-related terminology including “PID controller”, “robust control”, “adaptive control”, “model predictive control”, “optimal control”, and “LQR/LQG”. These were complemented by domain-specific descriptors, including “power systems”, “autonomous systems”, “cyber–physical systems”, “industrial control”, “robotics”, and “biomedical control”, as well as emerging technological terms such as “reinforcement learning”, “deep learning”, “digital twins”, “edge computing”, and “quantum control”.

Previous terms were grouped using OR operators within each category and combined across categories using AND operators to construct expressive and targeted Boolean queries. Search expressions were adjusted to each platform’s syntax to maximize recall while minimizing irrelevant matches. Only peer-reviewed studies and authoritative books were considered, and non-archival material, such as theses, technical reports, or preprints, was excluded to preserve methodological rigor.

The identification stage yielded 460 records. After deduplication and removal of clearly irrelevant entries based on title and abstract relevance, 380 studies advanced to screening. Screening followed the two-phase PRISMA protocol. Titles and abstracts were reviewed to determine whether each work addressed system modeling, observer theory, control-loop design, or their integration with modern CPS and AI-driven frameworks. Studies focused solely on general machine learning, analytics pipelines, or peripheral data-centric workflows without explicit control-theoretic grounding were excluded. The remaining studies underwent full-text eligibility assessment to confirm the presence of mathematically rigorous formulations, explicit links to stability, observability, or estimation theory, and analytical or empirical insights relevant to control practice. This process was independently performed by two members of the author team, and discrepancies were resolved through discussion. Inter-rater agreement was high (Cohen’s $k \approx 0.80$), reflecting high consistency throughout screening and eligibility evaluation.

Following final inclusion checks, 314 peer-reviewed studies and authoritative books were retained. A PRISMA-style flow diagram (Fig. 1) summarizes the identification, screening, exclusion, and inclusion stages, providing a transparent overview of the selection methodology.

Each included work was examined using a structured coding protocol designed to capture its contribution to the broader control-theoretic landscape. Particular attention was given to the following:

- The nature of the modeling formulation, including ODE-based models, descriptor and DAE representations, transfer-function and state-space descriptions, and hybrid or reduced-order approaches.
- Observer structures were analyzed across Kalman-based estimators, high-gain and extended high-gain observers, Laguerre polynomial-based designs, and hybrid or learning-enhanced estimation schemes.
- Controller architectures were assessed across PID variants, nonlinear control strategies, optimal and predictive frameworks, and robust and adaptive formulations, along with their corresponding application domains and system-level considerations, including robustness, scalability, resilience, real-time feasibility, and deployment constraints in CPS and industrial environments.

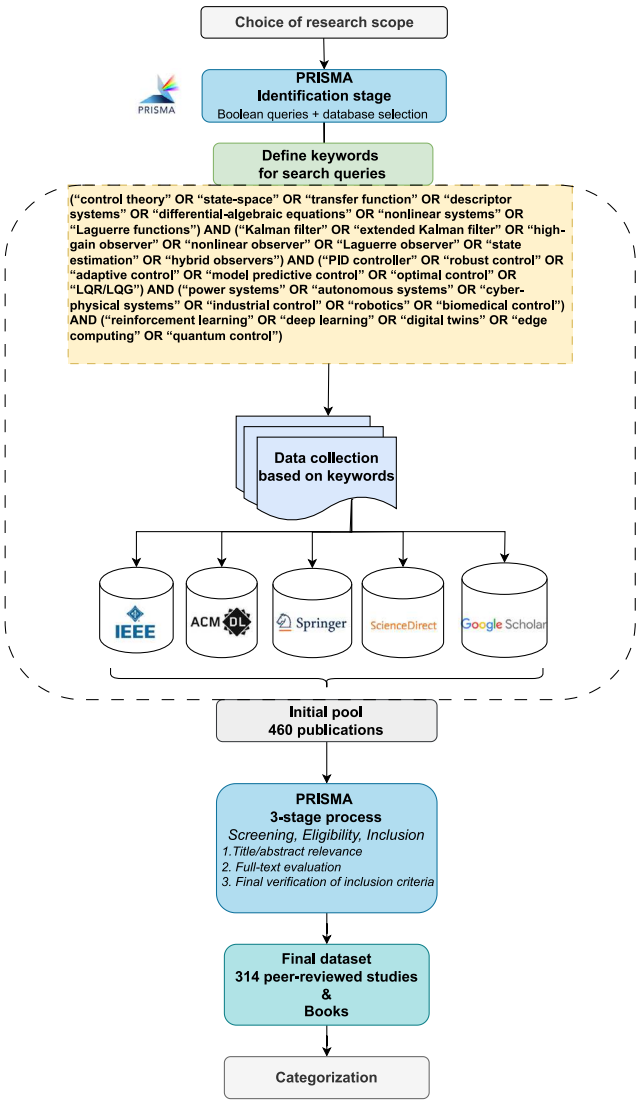


Fig. 1. PRISMA-Based Search and Selection Workflow for the Survey Studies.

This coding framework provides the analytical foundation for the integrative synthesis developed in the subsequent sections and supports a comprehensive understanding of how classical control theory and emerging intelligent paradigms jointly shape the design and operation of modern control systems.

1.3. Contribution

This survey makes four primary contributions to the literature on control theory and intelligent systems. They are organized thematically, and where appropriate, contrasted with prior research to clarify and highlight their added value.

- **Historical organization of classical control foundations.** The survey provides a structured overview of the historical foundations of control theory by organizing classical methods, systems modeling, behavior analysis, and PID control, together with their main modern extensions. This background contextualizes how established techniques connect to contemporary applications in domains such as power grids and cyber–physical systems, providing a coherent basis for the more advanced contributions that follow.
- **Nonlinear observer design within a unified estimation framework.** This survey consolidates nonlinear observer design into a unified analytical perspective by treating observer dynamics as a bridge between traditional model-based estimation and emerging data-driven methodologies. Particular emphasis is placed on observer formulations derived from Laguerre polynomial expansions, which are analyzed alongside EKFs, HGOs, and hybrid estimation schemes. This comparative and mathematically grounded treatment clarifies how Laguerre-based observers extend classical estimation tools through basis-function expansions, positioning them as an intermediate framework between traditional observer design and more data-driven estimation approaches—a connection that has not been explicitly articulated in prior surveys.
- **Cross-domain synthesis of control and AI paradigms.** This survey organizes the growing interaction between control theory and AI into a coherent cross-domain perspective, distinguishing learning-based components embedded directly within control loops from AI methods operating in adjacent areas such as robotics, computer vision, and data-driven perception. By examining how supervised learning, RL, and deep neural models interface with core control concepts, such as stability, observability, and feedback design, the survey provides a structured account of where AI augments, complements, or extends control-theoretic mechanisms. This synthesis clarifies how ML techniques can be incorporated into control systems with attention to theoretical grounding and interpretability, addressing a gap in previous interdisciplinary surveys that often treat control and AI developments in isolation.
- **System-level deployment and governance perspective.** This survey extends the discussion of control systems beyond methodological development by examining how scalability, resilience, cybersecurity, and governance requirements shape deployments in domains such as energy networks, transportation infrastructures, and cyber–physical systems. It highlights how practical constraints, such as communication bandwidth limits, latency, and packet drops in networked control, regulatory obligations, and safety-certification standards, directly influence the feasibility and adoption of advanced control and AI-driven strategies. By linking methodological advances with these system-level conditions, the survey provides a forward-looking perspective on how intelligent control architectures can be deployed sustainably, securely, and responsibly in large-scale operational environments.

Fig. 2 provides a structured overview of the key topics addressed in this study. The diagram positions the fundamentals of control theory and system modeling techniques at the foundation, capturing classical pillars such as PID controllers, state-space representation, stability analysis, and frequency-domain methods. Building on these foundations, the framework advances towards advanced nonlinear observer design, with an emphasis on descriptor systems, DAE observers, and Laguerre polynomial-based approaches. The middle layer highlights the applications of control theory in energy, transportation, healthcare, industrial IoT, and robotics, linking theory to domain practice. At the integration frontier, emerging technologies, including AI/ML, CPS, IoT, digital twins, quantum computing, and edge computing, are enablers of adaptive and intelligent control. The figure culminates in the identification of challenges and future trends, such as managing complexity, ensuring resilience, ethical governance, and bridging theory with real-world deployment. Together, these interconnected layers illustrate the survey's holistic perspective, spanning classical methods, application contexts, technological integration, and forward-looking directions in intelligent control systems.

The remainder of this paper is organized as follows: Section 2 presents the foundational concepts that underpin system modeling and dynamics analysis. Building on this basis, Section 3 explores the core

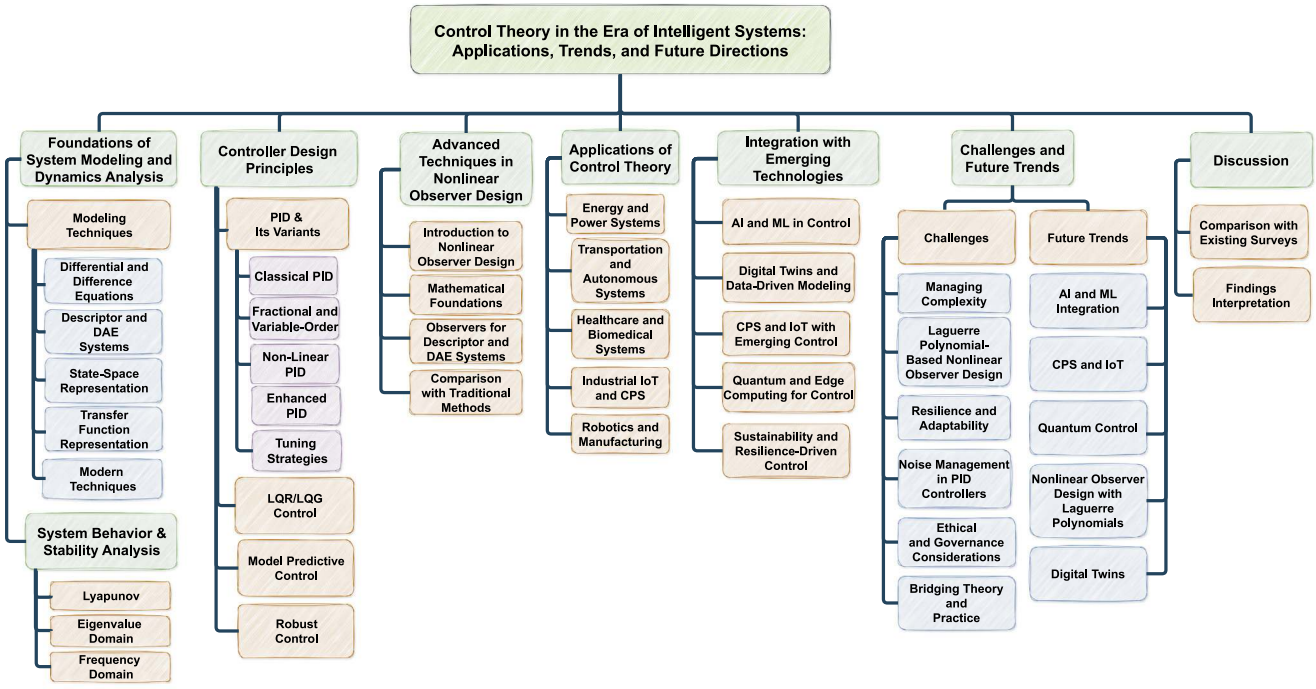


Fig. 2. Surveyed key topics in Control Theory in the Era of Intelligent Systems: Applications, Trends, and Future Directions.

principles of controller design, linking these foundations to practical regulation strategies. Next, Section 4 delves into advanced techniques in nonlinear observer design, with particular emphasis on methods that utilize Laguerre polynomials. Section 5 examines practical applications across various domains, illustrating the real-world relevance of theoretical constructs. Section 6 explores the integration of control systems with emerging technologies. Section 7 discusses future trends and challenges, highlighting the areas for ongoing research and development. Subsequently, Section 8 provides a comprehensive discussion synthesizing insights from the preceding sections. Finally, Section 9 concludes the survey and summarizes the key findings and their implications.

2. Foundations of system modeling and dynamics analysis

System modeling provides the analytical foundation for describing and predicting the behavior of dynamic processes. This section introduces the essential principles that govern model formulation and dynamics analysis, emphasizing how mathematical representations capture the physical laws, constraints, and interactions that shape system evolution. Accurate modeling is critical for assessing stability, transient behavior, and sensitivity to disturbances or parameter variations across engineering applications (Bala et al., 2017). The subsections that follow present the main modeling frameworks used in control-oriented analysis. These models provide the basis for subsequent analysis in Section 3, which introduces the principles of controller design and their role within modern feedback architectures.

2.1. Modeling techniques

Traditional system modeling relies primarily on differential and difference equations, transfer functions, and state-space models, which provide explicit and well-structured representations of system dynamics. Beyond these classical formulations, more advanced representations such as descriptor systems and DAEs extend the modeling framework to encompass algebraic constraints, multi-domain interactions, and structural couplings that arise in interconnected or large-scale processes.

A comparison of traditional modeling approaches in analog (or continuous-time) (Dorf & Bishop, 2016; Nise, 2020) and digital (or discrete-time) (Fadali & Visioli, 2009; Franklin et al., 1998; Isermann, 2013; Istepanian & Whidborne, 2012) domains is summarized in Table 2. These representations provide deterministic, interpretable descriptions of system behavior, offering precision and analytical clarity for linear, time-invariant processes. Nevertheless, real-world systems often exhibit nonlinearities, time-varying properties, structural constraints, or hybrid interactions that challenge the assumptions underlying these classical modeling frameworks. These limitations have driven the evolution of system modeling techniques toward more flexible modern methods capable of capturing and analyzing system dynamics, as the ability to do so accurately remains a cornerstone of effective control (Kulakowski et al., 2007; Lyshevski, 2012).

2.1.1. Differential and difference equations

Both analog and digital models can accommodate linear or nonlinear behavior, time-invariant or time-varying dynamics, and deterministic or stochastic inputs. The ability to model these systems using differential and difference equations provides a robust framework for simulation, stability analysis, and control design, bridging the gap between theoretical study and real-world applications in fields ranging from robotics to telecommunications. Using these models, engineers can design controllers that ensure stability, optimize performance, and adapt to changing conditions, regardless of whether the system operates in continuous or discrete time (Shinners, 1998).

Dynamic system modeling using differential equations forms the foundation of control theory by describing how systems evolve owing to internal dynamics and, in response, to external inputs and initial conditions. For analog systems, continuous-time models employ ODEs to capture the relationships among system states, inputs, and outputs. Such models precisely describe system dynamics in the time domain, enabling the analysis and control design of physical processes such as mechanical motion, thermal systems, and fluid dynamics. A typical LTI system is described as

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \tag{1}$$

Table 2
Comparison of traditional modeling approaches in analog and digital systems.

System type	Method	Description	Applications
Analog	Differential Equations	Continuous-time representation of state changes using ODEs.	Mass-spring-damper systems, electrical circuits.
	Transfer Functions	Input–output relationship represented in the frequency domain using the Laplace Transform.	Stability and dynamic response analysis in power systems.
	State-Space Representation	Compact representation using first-order differential equations for multivariable systems.	Observer design, pole placement, robotics.
	Descriptor/DAEs	Models combining differential and algebraic components with possibly singular matrix E; capture constraints and interconnections not reducible to ODE form.	Power networks (Kirchhoff's laws), constrained mechanical systems, and large-scale CPS.
Digital	Difference Equations	Discrete-time counterparts of differential equations, capturing state changes at discrete intervals.	Embedded system control, digital signal processing.
	Discrete Transfer Functions	Frequency-domain analysis for digital systems using z-domain representations.	Digital filter design, real-time control algorithms.
	Discrete State-Space Representation	Extends state-space models to discrete systems, defined through difference equations and equivalent matrix representations.	Real-time control in robotics and signal processing.
	Descriptor Difference Equations	They result from sampling (discretization) continuous-time descriptor systems and capture structural or physical couplings not representable by standard difference-equation or discrete state-space models. They combine dynamic updates with algebraic constraints.	Sampled constrained mechanical systems, digital control of power networks, and hybrid CPSs with algebraic coupling.

where $\mathbf{x}(t)$ is the state vector representing the system's internal variables, $\mathbf{A}$ is the state matrix dictating system dynamics, $\mathbf{B}$ is the input matrix mapping the influence of control inputs, and $\mathbf{u}(t)$ is the control input vector (Hinrichsen & Pritchard, 2005; Lunze, 2014).

In digital systems, discrete-time models represent dynamics as difference equations, where state transitions occur at discrete intervals. These systems are typically obtained by discretizing continuous models using methods such as zero-order hold or bilinear transformation. The discrete equivalent of the above equation is

$$\mathbf{x}[k+1] = \mathbf{A}_d\mathbf{x}[k] + \mathbf{B}_d\mathbf{u}[k], \tag{2}$$

where k represents the discrete time step, $\mathbf{A}_d$ and $\mathbf{B}_d$ are the discrete-time counterparts of $\mathbf{A}$ and $\mathbf{B}$, and $\mathbf{x}[k]$ and $\mathbf{u}[k]$ are the state and input vectors at step k. Discrete-time models are essential for systems in which computations are implemented digitally, such as microcontrollers and embedded systems (Dai et al., 2012; Tröltzsch, 2010).

Nonlinear systems extend this framework by introducing nonlinear state-input mappings

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)), \qquad \mathbf{x}[k+1] = f_d(\mathbf{x}[k], \mathbf{u}[k]), \tag{3}$$

where $f(\cdot)$ and $f_d(\cdot)$ denote the continuous- and discrete-time nonlinear dynamics, respectively. These equations account for complex behaviors such as saturation, hysteresis, and chaotic dynamics. Accurate modeling of such systems enables precise analysis, simulation, and control design for applications that span analog and digital environments (Chernous' ko et al., 2008).

2.1.2. Descriptor and DAE systems

Beyond conventional ODE- or difference equation models, many physical processes are naturally described by descriptor systems (DAEs), where algebraic constraints coexist with differential dynamics. A general linear DAE takes the form

$$\mathbf{E}\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \tag{4}$$

with a singular (non-invertible) matrix $\mathbf{E}$. Such formulations capture conservation laws, constraints, and interconnection structures that cannot be reduced to dynamic terms. In digital implementations, their counterparts appear as descriptor difference equations

$$\mathbf{E}_d\mathbf{x}[k+1] = \mathbf{A}_d\mathbf{x}[k] + \mathbf{B}_d\mathbf{u}[k]. \tag{5}$$

The analysis of controllability, observability, and stability requires generalized eigenvalue methods and index-reduction techniques that extend beyond standard LTI tools. Descriptor systems play an increasingly important role in large-scale networked CPS, where algebraic constraints, such as Kirchhoff's laws in power networks, must be preserved (Kunkel & Mehrmann, 2006; Kunkel et al., 2001).

2.1.3. State-space representation

The *state-space representation* is a mathematical framework for modeling, analyzing, and controlling dynamic systems, particularly those with multiple inputs and outputs. It describes system behavior using first-order differential or difference equations for discrete systems, providing a compact, comprehensive representation. Building on the formulation in (1), the state-space form is expressed as

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t), \tag{6}$$

where $\mathbf{x}(t)$ is the state vector representing the internal state of the system, $\mathbf{u}(t)$ is the input vector, $\mathbf{y}(t)$ is the output vector, $\mathbf{C}$ and $\mathbf{D}$ are matrices defining the system dynamics and output relations (Friedland, 2012).

The *state matrix* $\mathbf{A}$ captures the interconnections between the state variables, whereas $\mathbf{B}$ maps the inputs to states. The *output matrix* $\mathbf{C}$ relates the states to measurable outputs, and $\mathbf{D}$ describes the direct input–output relationships. This representation is advantageous for handling multivariable systems and is particularly useful in modern control techniques, such as pole placement and observer design (Jiang & Schaufelberger, 2005).

Key concepts such as *controllability* and *observability* are easily analyzed using state-space models (Montanari et al., 2023; Neto et al., 2022). A system is controllable if the state $\mathbf{x}(t)$ can be driven to any desired value using the input $\mathbf{u}(t)$. Mathematically, this can be verified using the controllability matrix

$$\mathbf{C} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}. \tag{7}$$

Similarly, a system is observable if the initial state $x(0)$ can be determined from the output measurements over time, as verified by the observability matrix:

$$\mathbf{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix}. \tag{8}$$

In digital systems, building on the formulation in (2), the dynamics are represented in discrete time as

$$\mathbf{y}[k] = \mathbf{C}_d\mathbf{x}[k] + \mathbf{D}_d\mathbf{u}[k], \tag{9}$$

where k is the discrete time step, and $\mathbf{C}_d, \mathbf{D}_d$ describe the discrete-time equivalent of the continuous system matrices.

The discrete state-space form is typically derived from a continuous-time representation using discretization methods, such as zero-order hold or bilinear (Tustin) transformation. These methods approximate the continuous-time behavior while preserving stability and system properties.

Controllability and *observability* are equally applicable to digital systems (Talapiden et al., 2024; Zhang et al., 2020). Observability is a structural property of control systems that is applicable to both linear and nonlinear systems. For nonlinear systems, advanced mathematical techniques from differential geometry, such as the Lie derivative and Lie brackets, are employed to analyze and establish observability conditions, extending the concept beyond the observability matrix used in linear systems (Van Handel, 2009).

Digital state-space representations are crucial for implementing control algorithms in real-time embedded systems, where computations are performed at discrete time intervals. They enable the design of digital controllers, such as linear quadratic regulators (LQRs) and linear quadratic Gaussian (LQGs) (Maghfiroh et al., 2022), as well as state observers, ensuring efficient, accurate, and stable control across robotics, signal processing, and industrial automation. The state-space approach supports modern computational methods for simulation, control design, and stability analysis, making it indispensable for complex, high-dimensional systems in fields such as aerospace, robotics, and power systems (Starr & Starr, 2020).

2.1.4. Transfer function representation

The transfer function representation is a cornerstone of control theory, providing a concise mathematical framework for analyzing and designing LTI dynamic systems in the frequency domain.

For *analog systems*, the transfer function is derived from the Laplace transform of the input–output relationship of the system, assuming zero initial conditions. It is defined as

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0}, \tag{10}$$

where $Y(s)$ and $U(s)$ are the Laplace transforms of the output and input signals, respectively, and a_i and b_i are the coefficients of the numerator and denominator polynomials, respectively. The poles, defined as the roots of the denominator, and zeros, defined as the roots of the numerator, play a critical role in determining the stability and transient response of the system (Snopek, 2024; Sundararajan, 2022a).

For *digital systems*, the transfer function is expressed in the z-domain using the Z-transform (Rao, 2018). The relationship is

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_m z^m + b_{m-1}z^{m-1} + \cdots + b_0}{a_n z^n + a_{n-1}z^{n-1} + \cdots + a_0}, \tag{11}$$

where z represents a discrete-time complex frequency variable. The poles and zeros in the z-domain determine the system behavior in discrete time. Stability in digital systems requires that all poles lie within the unit circle in the z-plane (Gazi, 2022; Sundararajan, 2022b).

Transfer functions facilitate the application of powerful tools, such as Bode plots, Nyquist diagrams, and root locus techniques, for controller design. By bridging the time and frequency domains, the transfer function representation is indispensable for understanding how systems respond to various inputs and for ensuring stability and performance in analog and digital systems (Leipholz, 2013; Rosenwasser & Lampe, 2006).

2.1.5. Modern modeling methods

Beyond traditional analytical and state-space models, contemporary system modeling approaches have evolved to meet the complexity of modern cyber–physical environments, enabling accurate system representation even when first-principles descriptions are incomplete.

In classical system modeling, the dynamics are typically expressed through first-principles formulations, $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta})$, where $\mathbf{x}(t)$ denotes the state vector, $\mathbf{u}(t)$ the control input, and $\boldsymbol{\theta}$ a set of physical parameters. While this approach ensures interpretability and physical consistency, it becomes impractical when the system's internal mechanisms are only partially known or too complex to capture analytically.

To overcome these limitations, *data-driven modeling* leverages experimental or simulated data to approximate the system dynamics directly as $\hat{\mathbf{f}}(\mathbf{x}, \mathbf{u}) \approx \mathbf{f}(\mathbf{x}, \mathbf{u}, \boldsymbol{\theta})$, often using regression-based or neural architectures to represent nonlinear relationships between inputs and states (Brunton & Kutz, 2022). Within this paradigm, *system identification* and *ML* methods enable accurate representations even when theoretical derivation is infeasible, while *hybrid (gray-box) approaches* combine physical insight with empirical adaptability to balance interpretability and predictive accuracy (Bazanella et al., 2023; Engell et al., 2003).

Hybrid (gray-box) models merge physics-based and data-driven representations, where analytical dynamics are complemented by learned residual or scaling components. This formulation enables the model to preserve physical interpretability while capturing unmodeled or nonlinear behaviors from data. Widely adopted in process systems, control design, and physics-informed learning frameworks, these hybrid approaches bridge the gap between mechanistic modeling and machine learning. They enhance model accuracy and adaptability, making them particularly effective for complex CPSs where first-principles descriptions are incomplete or analytically intractable (Seyyedi et al., 2023).

Building on the same principle, digital-twin frameworks extend hybrid modeling into real-time operation by continuously coupling sensor data with virtual counterparts, enabling predictive monitoring and adaptive control across industrial, robotic, and energy applications (Rasheed et al., 2020).

Reduced-order and surrogate modeling methods also support implementations where real-time computation and energy efficiency are critical. Such approaches, including model order reduction for optimal control problems and projection-based techniques such as proper orthogonal decomposition (POD) or balanced truncation, preserve control-relevant dynamics while significantly reducing computational complexity (Hinze, 2023). These modern approaches complement classical modeling techniques by enhancing adaptability, interpretability, and scalability, thus bridging the gap between theoretical control design and practical deployment in intelligent, data-rich environments.

These modeling paradigms form the basis for analyzing the dynamic behavior of controlled systems. The next subsection builds on these representations to examine stability, transient dynamics, and frequency response—core properties that determine system robustness and dynamic performance.

2.2. System behavior and stability analysis

Lyapunov stability is a general method for analyzing the stability of both linear and nonlinear systems. The Lyapunov stability function $V(\mathbf{x})$ is a scalar function used to assess the stability of the equilibrium points in dynamic systems. It provides an energy-like measure, where the value of $V(\mathbf{x})$ reflects the system's "distance" from equilibrium, and its derivative determines the system's trajectory towards or away from equilibrium (Khalil, 2009).

For an analog (continuous time) system described by $\dot{\mathbf{x}}(t) = f(\mathbf{x}(t))$, the stability conditions are

$$V(\mathbf{x}) > 0 \quad \text{and} \quad \dot{V}(\mathbf{x}) = \frac{\partial V}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} < 0, \quad \forall \mathbf{x} \neq \mathbf{0}, \tag{12}$$

Table 3
Routh-Hurwitz table.

s^n	a_n	a_{n-2}	a_{n-4}	a_{n-6}	a_{n-8}	$\cdots$
s^{n-1}	a_{n-1}	a_{n-3}	a_{n-5}	a_{n-7}	a_{n-9}	$\cdots$
s^{n-2}	$b_1 = -\frac{a_n a_{n-3} - a_{n-1} a_{n-2}}{a_{n-1}}$	$b_2 = -\frac{a_n a_{n-5} - a_{n-1} a_{n-4}}{a_{n-1}}$	b_3	b_4	$\cdots$	
s^{n-3}	$c_1 = -\frac{a_{n-1} b_1 - b_1 a_{n-3}}{b_1}$	$c_2 = -\frac{a_{n-1} b_3 - b_1 a_{n-5}}{b_1}$	c_3	c_4	$\cdots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\cdots$	
s^0	a_0					

while for a digital (discrete time) system $\mathbf{x}[k+1] = f(\mathbf{x}[k])$, they are formulated as

$$V(\mathbf{x}) > 0 \quad \text{and} \quad V(\mathbf{x}[k+1]) - V(\mathbf{x}[k]) < 0, \quad \forall \mathbf{x} \neq \mathbf{0}. \tag{13}$$

These conditions ensure that the system converges to an equilibrium point. The principles of stability remain the same as in the continuous case, but the derivative $\dot{V}(\mathbf{x})$ is replaced by the difference $\Delta V(\mathbf{x})$.

Lyapunov stability is widely applied in power systems to maintain grid stability under dynamic conditions, in robotic systems to ensure stable trajectories and control mechanisms, in adaptive control, and in nonlinear dynamics, where classical eigenvalue methods are insufficient. This versatility makes Lyapunov's methods a cornerstone of modern control theory (Nguyen & Nguyen, 2018).

Eigenvalue analysis is a fundamental method for assessing the stability of LTI systems (de Almeida & Araújo, 2019). For a system represented in state-space form, either in analog $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$ or in digital $\mathbf{x}[k+1] = \mathbf{A}_d\mathbf{x}[k]$, stability is determined directly from the eigenvalues of the corresponding system matrices. In continuous-time systems, stability is guaranteed when all eigenvalues λ of $\mathbf{A}$ satisfy $\Re(\lambda) < 0$ (all eigenvalues lie in the left half of the complex plane), ensuring exponential decay of system trajectories. In discrete-time systems, stability requires that all eigenvalues λ_d of $\mathbf{A}_d$ lie strictly inside the unit circle, i.e., $|\lambda_d| < 1$.

Eigenvalue-based stability analysis is widely used in aerospace, electrical circuit modeling, and general control engineering, where state-space formulations are the standard representation (Lin et al., 2020; Mehrmann & Voss, 2004).

The *Routh–Hurwitz criterion* (Bodson, 2020; Patil, 2021) is an algebraic method for determining the stability of an LTI system by analyzing the coefficients of its characteristic equation, $P(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$. For the system to be stable, all roots of $P(s)$ must have negative genuine parts, which means that they must lie strictly in the left half-plane. Instead of explicitly computing the roots, the Routh–Hurwitz criterion determines stability by constructing the Routh array, a structured tabular arrangement of the polynomial coefficients.

The first row of the array consists of the coefficients of $s^n, s^{n-2}, s^{n-4}, \ldots$, whereas the second row consists of the coefficients of $s^{n-1}, s^{n-3}, s^{n-5}, \ldots$. Each subsequent row is derived recursively from the preceding two rows using a determinant-based relationship. Specifically, the Routh table is computed as shown in Table 3. This process continues until all the rows are completed. For the system to be stable, all the elements in the first column of the Routh array must be strictly positive. Any sign change in this column indicates the number of roots with positive real parts, which corresponds to the number of unstable poles. If a zero appears in this column, special handling, such as the epsilon method or differentiation of the characteristic equation, is required to proceed with the stability analysis.

For discrete-time systems, the stability criterion differs because the poles must be located inside the unit circle. The Jury stability test is used to verify that all roots of the characteristic polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0$ lie strictly within $|z| < 1$. This method involves a sequence of algebraic conditions applied to the polynomial coefficients, making it a suitable counterpart to the Routh–Hurwitz test for continuous-time systems (Bodson, 2020; Ho et al., 1998).

In control theory, *frequency-domain* analysis provides powerful tools for understanding and designing dynamic systems by examining their behavior across a spectrum of frequencies. Unlike time-domain methods, which focus on system behavior over time, frequency-domain techniques analyze the response of systems to sinusoidal inputs of varying frequencies. This approach is beneficial for characterizing the system stability, robustness, and performance (Jin et al., 2019; Tian & Gao, 2007).

Key methods include the Nyquist criterion, which determines closed-loop stability by evaluating how the open-loop transfer function encircles the critical point $-1 + j0$ in the complex plane, and the Bode plot, which graphically represents a system's gain (magnitude) and phase shift as a function of frequency. These tools provide intuitive insights into the phase and gain margins, which are critical metrics that quantify a system's resilience to parameter variations or disturbances (Desoer & Wang, 1980; Liao & Wang, 2018).

Frequency-domain analysis also supports filter design, resonance avoidance, and noise rejection by enabling the precise tuning of system responses at specific frequencies. Widely applied in fields such as signal processing, communications, and control system design, frequency-domain methods remain indispensable for achieving the desired performance and stability in both linear and nonlinear systems (Boiko, 2008; Seron et al., 2012).

With these system-level insights established, we now turn to the principles that guide the design of controllers capable of shaping the resulting dynamics.

3. Controller design principles

Controllers determine how a dynamic system reacts to reference commands, disturbances, and uncertainties. Within a feedback loop, corrective actions are generated by continuously comparing the system output to the desired reference, a mechanism that is fundamental to achieving stability, robustness, and accurate tracking performance (Ozbay, 2019). As illustrated in Fig. 3, the controller computes the control input $u(t)$ based on the reference signal $r(t)$ and the estimated state vector $x(t)$ provided by the observer. The observer reconstructs the internal states from the measured output $y(t)$, forming a closed interaction between estimation and control that underlies both classical and modern feedback architectures.

Classical feedback strategies, most notably PID controllers, represent the earliest and most widely implemented control approaches due to their simplicity and effectiveness in regulating linear, single-loop processes (Díaz-Rodríguez et al., 2019; Rodic, 2009). As engineering systems grew in complexity, state-space formulations introduced a systematic framework for multivariable and nonlinear control, enabling structured analysis and design. Within this framework, LQR and its stochastic counterpart, the LQG controller, introduced optimal quadratic-performance design with Kalman filtering, forming the basis for many observer-based multivariable control schemes. Building upon this foundation, adaptive and robust control expanded closed-loop capabilities by allowing online parameter adjustment or guaranteeing performance under modeling uncertainties and external disturbances (Liu & Yao, 2016; Narendra, 2012; Tsui, 2022).

Recent developments, including MPC and data-driven control strategies, leverage optimization and computational intelligence to anticipate future system behavior, incorporate constraints, and adapt to changing operating conditions. These advances reflect a transition from reactive to predictive and learning-enhanced control paradigms, supporting demanding applications in autonomous systems, renewable energy, and large-scale cyber–physical infrastructures (Azar & Vaidyanathan, 2015; Williams et al., 2018).

The remainder of this section provides a structured overview of these controller families, outlining their fundamental principles, design methodologies, and relevance to contemporary engineering applications.

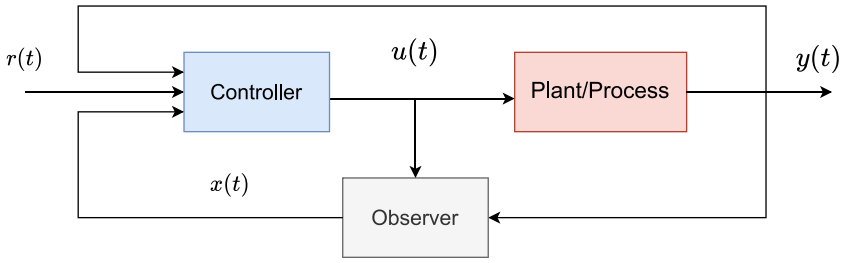


Fig. 3. General control-observation framework illustrating the closed-loop interaction among the controller, plant, and observer subsystems.

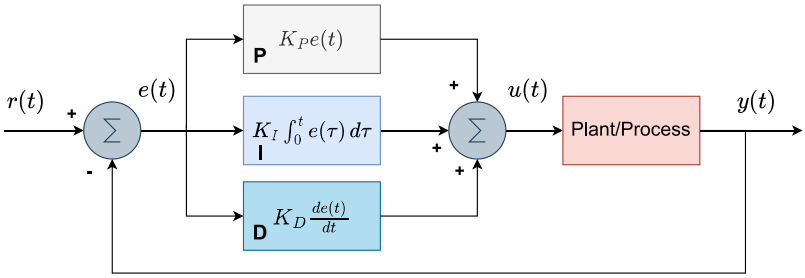


Fig. 4. Block diagram of the PID control loop showing error computation, proportional–integral–derivative actions, and feedback correction.

3.1. PID and its variants

PID controllers remain the most widely used feedback control strategy across process industries, robotics, and general automation, owing to their simplicity, robustness, and reliable performance in a broad range of operating conditions.

3.1.1. Classical PID control

Classical PID controllers operate by transforming the tracking error into a corrective control action $u(t)$ composed of three terms: the proportional term K_P, which reacts to the instantaneous error, the integral term K_I, which accumulates past errors to eliminate steady-state offsets, and the derivative term K_D, which estimates the error's rate of change to improve damping and transient response. Their functional structure is illustrated in Fig. 4.

At each control cycle, the plant (actual system) output $y(t)$ is measured and compared with the reference/desired $r(t)$, generating the error $e(t) = r(t) - y(t)$. The proportional, integral, and derivative contributions are evaluated and combined to produce the control input $u(t)$, which is then applied to the plant, forming the closed feedback loop. The resulting control action is mathematically expressed as

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}. \tag{14}$$

In practical implementations, the derivative term ($K_D \frac{de(t)}{dt}$) is typically filtered using a low-pass filter to attenuate high-frequency noise that could destabilize the system. This filtered derivative is crucial for real-world performance and ensures robust operation (Rojas et al., 2021; Vagia, 2012).

PID controllers are particularly effective for linear systems, where they ensure stable and accurate performance by dynamically adjusting the control output to reduce the error and improve the response time. Despite their simplicity, they often require careful gain tuning to achieve optimal performance, particularly in systems with complex dynamics (Somefun et al., 2021).

In digital implementations, the continuous-time control law must be executed through discrete computations at successive sampling instants, often combined with derivative filtering and anti-windup protection to ensure robust operation. In discrete systems, PID controllers operate on signals sampled at fixed time intervals, allowing the control law to be executed by microcontrollers or digital processors. The continuous-time PID expression is therefore translated into a discrete-time form, where the control input at step k is given by

$$u[k] = K_P e[k] + K_I \sum_{i=0}^{k} e[i]\Delta t + K_D \frac{e[k] - e[k-1]}{\Delta t}, \tag{15}$$

where $e[k]$ is the error at the kth time step, defined as $e[k] = r[k] - y[k]$, $r[k]$ is the desired setpoint, $y[k]$ is the actual output and Δt is the sampling interval (Eriksson & Koivo, 2005; Merrikh-Bayat et al., 2015).

In addition to the conventional discrete PID formulation, an alternative incremental (or velocity) form of the PID controller has been widely adopted in digital-control implementations. This version expresses the control input based on the differences in errors, thereby improving memory efficiency and mitigating integral windup. The incremental form is given by

$$u[k] = K_P e[k] + K_I \Delta e[k] + K_D \Delta^2 e[k], \tag{16}$$

where $\Delta e[k] = e[k] - e[k-1]$ and $\Delta^2 e[k] = e[k] - 2e[k-1] + e[k-2]$. A recent study Rospawan et al. (2024) highlighted the advantages of this form for adaptive and efficient PID control in real-time digital systems, particularly in multi-input multi-output (MIMO) processes.

Beyond their classical formulation, discrete PID controllers are well aligned with modern digital platforms, where the control law is executed at fixed sampling instants. Their structure allows efficient implementation within embedded and processor-based architectures, supporting a wide range of real-time applications (Chen & Ricles, 2008; Corsi et al., 2017).

However, discrete implementations are not without their challenges. One critical factor is the choice of the sampling interval (Δt), which directly affects the accuracy and stability of the controller. Furthermore, numerical precision limitations in digital systems, often referred to as quantization errors, can introduce minor computational inaccuracies in the results. Despite these challenges, discrete PID controllers remain fundamental to digital control systems, delivering efficient and reliable performance in applications such as robotics, industrial automation, and embedded systems (Laskawski & Wcislik, 2016; Wang et al., 2021).

The formulation discussed above corresponds to a conventional integer-order PID controller, where both the integral and derivative actions are of first order. This serves as the foundational structure upon which more generalized forms, such as fractional- and variable-order PID controllers, have been developed, as discussed in the following section.

3.1.2. Fractional and variable-order PID control

While conventional PID controllers rely on integer-order differentiation and integration, recent advances have introduced fractional-order PID (FOPID) and variable-order PID (VOPID) controllers, which generalize the classical structure and enhance flexibility in shaping the system response. A typical FOPID controller is formulated as

$$u(t) = K_P e(t) + K_I D^{-\lambda} e(t) + K_D D^{\mu} e(t), \tag{17}$$

where $D^{-\lambda}$ and D^{μ} denote the fractional integral and derivative operators of non-integer orders λ and μ, respectively. By tuning these orders together with the proportional, integral, and derivative gains, FOPID controllers allow finer adjustment of the transient and steady-state characteristics, providing superior robustness, disturbance rejection, and noise attenuation compared with their integer-order counterparts (Monje et al., 2010; Soukkou et al., 2016).

The block structure of a FOPID remains identical to that of the conventional PID controller, as illustrated in Fig. 5, with the only difference being that the integral and derivative actions are replaced by fractional operators $D^{-\lambda}$ and D^{μ}. This structural consistency facilitates implementation while expanding the controller's degrees of freedom through fractional calculus.

VOPID controllers further extend this concept by allowing the fractional orders to vary dynamically with time or system conditions, expressed as $\lambda = \lambda(t)$ and $\mu = \mu(t)$. Such adaptation enables real-time reshaping of the control law to match evolving plant dynamics, yielding improved performance under nonlinearities, parameter drifts, and operating uncertainties. Although their schematic representation is identical to that of the FOPID, the differentiation and integration blocks are variable-order operators that are updated online via adaptive mechanisms or fuzzy/neural rules. These controllers have demonstrated strong potential in energy management systems, biomedical regulation, and robotics, where dynamic adaptation is crucial (Chen et al., 2020).

Both FOPID and VOPID controllers benefit from modern digital implementations, where efficient numerical approximations of fractional and variable-order operators enable real-time computation on embedded and programmable logic controller (PLC)-based hardware (Możaryn et al., 2021; Oprzędkiewicz et al., 2021).

While fractional and variable-order formulations generalize the internal structure of the PID controller by modifying the order of the integral and derivative actions, they still operate within a fundamentally linear framework. In contrast, nonlinear PID (N-PID) controllers introduce explicit nonlinear transformations of the error or its components, offering an alternative means of enhancing performance in systems where linear control action is insufficient.

3.1.3. Non-linear PID control

N-PID controllers (Sun et al., 2022) retain the classical PID structure but introduce nonlinear transformations of the error or the gain terms. These modifications do not make the controller inherently nonlinear; rather, they enhance robustness, reduce overshoot, and improve control authority when regulating nonlinear or time-varying plants.

Typical approaches (see Table 4) include trigonometric nonlinearities such as hyperbolic functions that amplify small errors while suppressing large ones, power-law mappings that reshape the error dynamics through fractional or bounded exponents, and saturation- or exponential-based gain functions that prevent excessive control effort in high-error regimes. Other N-PID formulations employ gain scheduling, where the controller gains vary as functions of the error magnitude or system states, or introduce engine-type nonlinearities incorporating delayed or filtered error terms.

Table 4
Representative formulations of N-PID control laws (Çelik et al., 2026).

Control Law $u(t)$
$u(t) = K_P\,\alpha\big(e(t)\big)\,e(t) + K_I\,\beta\big(e(t)\big)\int_0^t e(\tau)\,d\tau + K_D\,\gamma\big(e(t)\big)\,\frac{de(t)}{dt}$
$u(t) = K_P\,f_P\big(e(t)\big) + K_I\,f_I\Big(\int_0^t e(\tau)\,d\tau\Big) + K_D\,f_D\Big(\frac{de(t)}{dt}\Big)$
$u(t) = K_P\,\tanh\big(e(t)\big) + K_I\,\mathrm{sat}\Big(\int_0^t e(\tau)\,d\tau\Big) + K_D\,\frac{\frac{de(t)}{dt}}{1+\left\lvert\frac{de(t)}{dt}\right\rvert}$

In practical implementations, the parameters of these nonlinear transformations are typically tuned through modified classical rules or via adaptive and optimization-based procedures that adjust the controller behavior to the underlying nonlinear dynamics. Such tuning mechanisms allow N-PID controllers to maintain stability and effectiveness across a wide range of operating conditions (Çelik et al., 2026).

3.1.4. Enhanced PID control

Enhanced PID control encompasses mechanisms and computational techniques that extend the capabilities of the classical PID structure. These approaches may either embed adaptive or rule-based elements, such as fuzzy inference, or employ optimization to improve controller performance under varying operating conditions.

A typical class of enhanced PID control relies on the iterative adjustment/update of the controller gains within an optimization loop governed by a performance index such as the integral absolute error (IAE), integral squared error (ISE), integral of time-weighted squared error (ITSE), or integral time-weighted error (ITAE). These criteria directly quantify transient and steady-state performance, allowing gains to be derived analytically or through numerical optimization. Nature-inspired/metaheuristic algorithms offer a global search capability and are widely used to adjust the PID gains when analytical or classical numerical methods are insufficient or difficult to apply. Techniques such as genetic algorithms, particle swarm optimization (PSO), and differential evolution explore the gain space efficiently and identify suitable parameter combinations that satisfy the selected performance criterion (Joseph et al., 2022; Patil et al., 2024). The overall workflow of this optimization-based update mechanism is illustrated in Fig. 6.

Beyond global optimization approaches, other strands of enhanced PID control focus on online gain adaptation. Fuzzy-logic–based schemes constitute a widely used framework in this category, enabling real-time adjustment of K_P, K_I, and K_D based on instantaneous error information. An illustrative example is shown in Fig. 7, where a fuzzy inference mechanism updates the gains according to $e(t)$ and $\frac{de(t)}{dt}$. Under nonlinear or time-varying dynamics, this additional inference layer refines the closed-loop response and enhances robustness. Recent adaptive variants extend this principle by incorporating online evaluation of operating conditions. For instance, Han et al. developed an adaptive fuzzy PID control strategy in which fuzzy rules and gain adjustments are continuously modulated using real-time road condition estimation and suspension performance feedback (Han et al., 2022).

Overall, these enhanced mechanisms expand the operating range and adaptability of the classical PID framework, but their effectiveness ultimately depends on how the underlying gains are selected and updated, a topic explored in the following section on tuning strategies.

3.1.5. Tuning strategies

PID controllers are widely used in industrial automation, where the selection of proportional, integral, and derivative gains influences the system performance.

The proportional gain affects response speed but may introduce overshoot. The integral gain helps reduce steady-state errors, and the

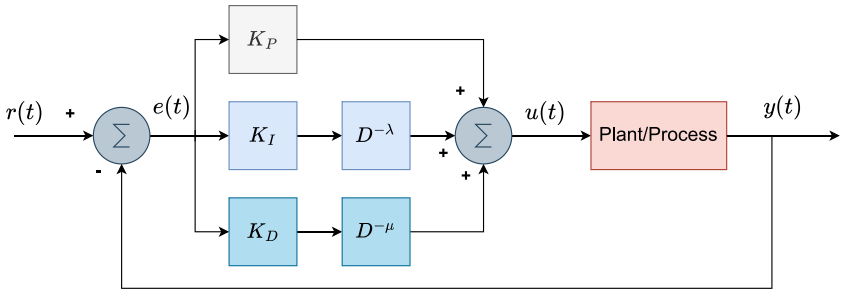


Fig. 5. Block diagram of fractional-order and variable-order PID controllers. In the VOPID, the fractional orders vary as $\lambda = \lambda(t)$ and $\mu = \mu(t)$.

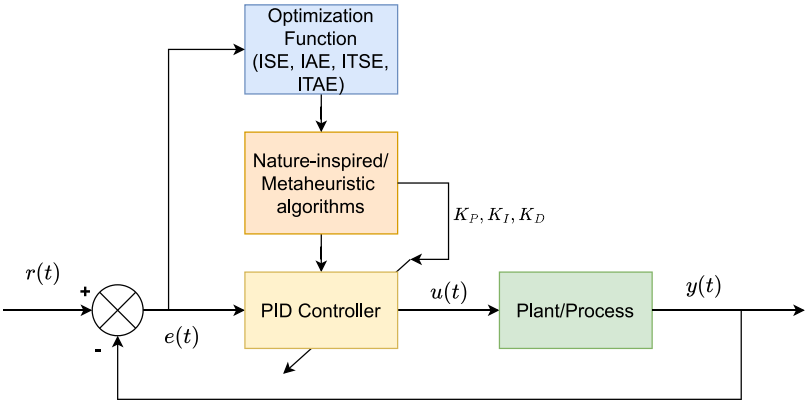


Fig. 6. Block diagram of Nature-inspired/Metaheuristic PID controller.

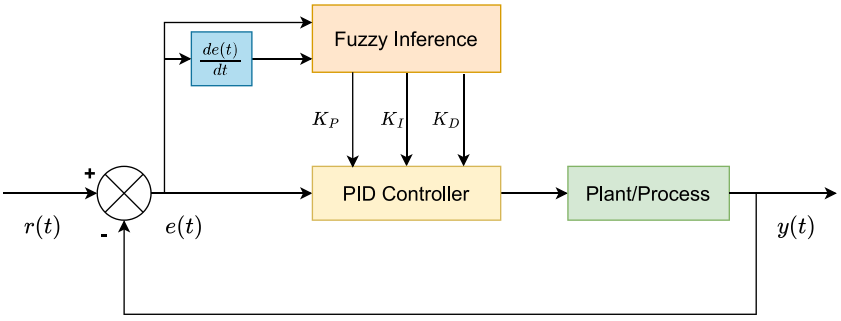


Fig. 7. Block diagram of Fuzzy Adaptive PID controller.

derivative gain improves stability by anticipating changes. The effectiveness of these parameters depends on system dynamics rather than being inherently tied to specific applications. Proper tuning strategies (Borase et al., 2021) are essential for achieving the desired control objectives in various industrial settings.

Classical empirical approaches, such as the Ziegler—Nichols tuning rules, determine these parameters experimentally by identifying the ultimate gain K_u and oscillation period T_u at the stability boundary of a proportional-only loop. The controller gains are then computed using empirical ratios: $K_P = 0.6K_u$, $T_I = 0.5T_u$, and $T_D = 0.125T_u$, established through extensive experimentation to yield a fast transient response with approximately 25% overshoot and acceptable stability margins across a broad class of linear processes. Although straightforward, these settings often lead to aggressive control actions when applied to nonlinear or delay-dominant systems.

Several alternative empirical schemes have extended the Ziegler–Nichols concept toward more conservative or model-aware formulations. The Cohen–Coon method uses a first-order-plus-dead-time (FOPTD) model, the delay time (L), time constant (T), and process gain (K) to derive closed-form tuning equations that explicitly account for process delay, resulting in smoother transient behavior. The Tyreus–Luyben rules modify Ziegler–Nichols constants to achieve greater phase margin and robustness, while the Chien–Hrones–Reswick method, a refined variant of the Ziegler–Nichols step-response approach, provides separate tuning tables for setpoint tracking and disturbance rejection. Furthermore, the Åström–Hägglund relay auto-tuning technique replaces manual gain adjustment with a self-oscillating relay experiment that automatically estimates K_u and T_u, thus enabling automated initialization of PID parameters (Singh et al., 2023). These schemes remain attractive for industrial deployment due to their low computational requirements and intuitive implementation (Borase et al., 2021).

As summarized in Table 5, classical empirical approaches remain attractive for rapid deployment due to their simplicity, yet they offer limited robustness under nonlinear or time-delay conditions. Model-based and optimization-driven schemes overcome these limitations by explicitly minimizing performance indices. In contrast, intelligent and data-driven methods enhance adaptability by leveraging real-time information and feedback-driven adjustment mechanisms.

Building on these data-driven strategies, ML-based PID tuning (Çelik et al., 2026) is a more specialized class in which regression models, neural networks (NNs), RL methods such as Q-learning, and recurrent

Table 5
Comparison of representative PID tuning strategies.

Method	Model/Information required	Design objective	Controller nature	Remarks/Typical use
Ziegler–Nichols	Ultimate gain K_u and period T_u from oscillation test	≈25% overshoot, fast response	Aggressive/Empirical	Simple experimental procedure; limited robustness for nonlinear or delay-dominant systems.
Cohen–Coon	FOPTD model (K, T, L)	Balanced transient and steady-state behavior	Semi-empirical	Accounts for process delay; smoother response for open-loop identification.
Tyreus–Luyben	Ultimate gain and period	Enhanced phase margin and robustness	Conservative/Empirical	Modified Ziegler–Nichols constants; improved stability for industrial loops.
Chien–Hrones–Reswick	Step response (rise and delay times)	Separate tuning for setpoint or disturbance rejection	Flexible/Empirical	Modified Ziegler–Nichols step response method;Different parameter sets for servo and regulatory control modes.
Åstr$\grave{o}$m–H$\acute{a}$gglund	Relay auto-oscillation test	Self-tuning without manual gain search	Adaptive/Experimental	Automated identification of K_u and T_u; suitable for embedded or online tuning.
Model-based Optimization	Process model or time-domain error metrics	Minimization of performance indices; frequency-domain constraints possible	Analytical/Numerical	Balances transient and steady-state behavior; extendable to loop-shaping or pole-placement design.
Metaheuristic Optimization	Performance index and search space definition	Global search for optimal gains under nonlinear dynamics	Intelligent/Global	Model-free optimization; handles nonconvex, uncertain, or multi-objective tuning problems.
Fuzzy Adaptive PID	Error signals and fuzzy rule base	Online gain adjustment according to error trends	Intelligent/Adaptive	Improves robustness and learning capability; effective for time-varying and nonlinear systems.
ML-based Tuning	Operational or historical process data	Learning-based estimation of gain adjustments using data-driven models	Intelligent/Data-driven	Captures nonlinear and time-varying dynamics; regression models, NNs, RL; suitable when explicit process modeling is difficult.

architectures infer gain updates directly from process data. Hybrid approaches further integrate ML with conventional control principles, providing improved robustness and scalability in nonlinear, uncertain, or high-dimensional settings.

These comparisons highlight the continuous evolution from static empirical tuning toward self-optimizing and adaptive PID paradigms.

Overall, the evolution of PID control has been shaped far more by advances in tuning methodologies than by modifications to the controller's basic structure. Modern tuning approaches improve transient response and disturbance handling, making PID controllers more effective even in complex or poorly modeled systems.

3.2. LQR and LQG controllers

Beyond classical feedback laws, optimal control introduces a formal framework that explicitly balances control performance and energy expenditure through quadratic cost minimization. Among these formulations, the LQR and LQG controllers represent foundational strategies in modern control theory (Kumar & Gupta, 2023).

The LQR provides the optimal full-state feedback gain $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^{\top}\mathbf{P}$ by minimizing a quadratic cost on state deviation and control effort, where $\mathbf{P}$ satisfies the algebraic Riccati equation. The matrix $\mathbf{R} \succ 0$ acts as a weighting factor that penalizes excessive control effort in the quadratic cost function. Its selection determines the trade-off between system performance and energy expenditure, influencing the aggressiveness of the control action. Smaller values of $\mathbf{R}$ yield faster but more energy-intensive responses, whereas larger values lead to smoother, more conservative control.

Since $\mathbf{R}$ is selected based on practical considerations rather than system physics, recent work highlights the need for systematic tuning methods (weighting-matrix selection methods) to ensure consistent performance across operating conditions (Chacko et al., 2024).

In practical deployments, full-state measurements are rarely available, motivating the LQG formulation, which combines the LQR with a Kalman state estimator (Tang et al., 2023). The estimator employs the gain $\mathbf{L} = \mathbf{P}_e\mathbf{C}^{\top}\mathbf{V}^{-1}$, obtained from the estimator Riccati equation

$$\mathbf{A}\mathbf{P}_e + \mathbf{P}_e\mathbf{A}^{\top} - \mathbf{P}_e\mathbf{C}^{\top}\mathbf{V}^{-1}\mathbf{C}\mathbf{P}_e + \mathbf{W} = 0, \tag{18}$$

to minimize the steady-state estimation error covariance under process and measurement noise covariances $\mathbf{W}$ and $\mathbf{V}$. The resulting observer–based control law $\mathbf{u}(t) = -\mathbf{K}\,\hat{\mathbf{x}}(t)$ uses the estimated states $\hat{\mathbf{x}}(t)$ instead of the true states $\mathbf{x}(t)$. Because of the separation principle, the controller gain $\mathbf{K}$ and the estimator gain $\mathbf{L}$ can be computed independently, providing an optimal, minimum-variance control structure that generalizes the classical state-feedback framework to noisy and partially observed systems.

Recent advances in risk-sensitive LQG reinterpret the classical estimator–controller structure through an exponential-cost formulation that is mathematically equivalent to mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control and linear–quadratic zero-sum games. This equivalence provides robustness to model mismatch and disturbances, a property explicitly demonstrated in recent analysis of small-disturbance input-to-state stability. Cui et al. (2024).

Beyond their classical formulation, nonlinear trajectory-optimization methods such as Iterative LQR (iLQR) and Differential Dynamic Programming (DDP) can be viewed as generalizations of the LQR principle to nonlinear systems, relying on iterative local approximations of the dynamics and cost along a nominal trajectory (Chen et al., 2019; Iwanaga & Fujimoto, 2025).

3.3. Model predictive control

MPC represents a class of optimization-based controllers that compute the control input by repeatedly solving a finite-horizon prediction problem. At each sampling instant, the controller predicts the future system behavior using an explicit process model and selects the control sequence that minimizes a cost function while satisfying input, state, and output constraints. Only the first element of this sequence is applied, and the process repeats at the next step, providing a receding-horizon implementation (Raković, 2016).

The general MPC structure integrates a prediction model with an online optimization routine, forming a closed-loop strategy that continuously updates based on the latest measurements. As shown in Fig. 8, the controller receives the reference or exogenous input r together with

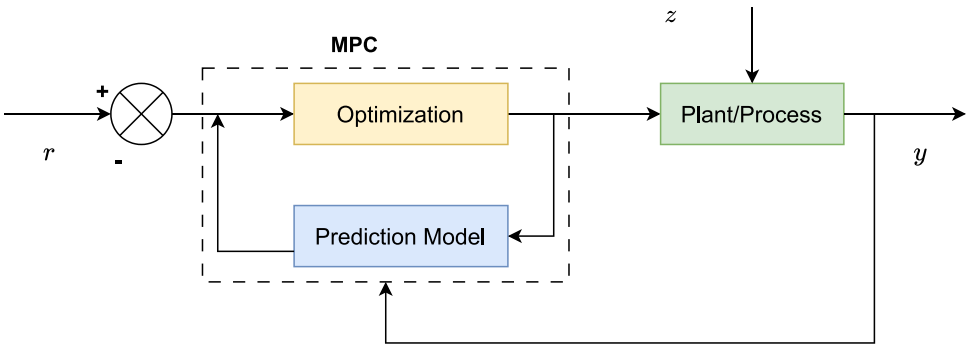


Fig. 8. Basic architecture of an MPC. The controller uses a prediction model and an online optimization routine to compute the constrained optimal input u, which is applied to the plant.

the measured output y to update its internal model and solve a constrained optimization problem. The resulting control input u is applied to the plant, while the predicted performance variables z are evaluated inside the cost and constraint set. This cyclic optimization-prediction mechanism enables MPC to adapt in real time to disturbances and modeling deviations.

A key advantage of MPC is its ability to incorporate multivariable interactions, hard constraints, and time-varying references directly into the control law, features that are difficult to handle using classical linear controllers. Modern formulations introduce robustness through tube-based and min–max MPC schemes, enabling stable operation under bounded disturbances and model uncertainties (Raghuraman & Koeln, 2021).

Beyond these formulations, MPC has evolved into advanced variants such as nonlinear (NMPC) (Nan et al., 2022), and stochastic (SMPC) (Mesbah, 2016), which extend the classical framework to accommodate system nonlinearities, and probabilistic constraints objectives without altering its fundamental structure.

MPC has therefore become one of the most widely adopted advanced control strategies in process industries, autonomous driving, energy systems, and embedded CPS implementations (Schwenzer et al., 2021).

3.4. Robust control

Robust control methods aim to guarantee stability and performance in the presence of model uncertainties, disturbances, and unmodeled dynamics. In contrast to classical controllers that rely on nominal system representations, robust formulations explicitly incorporate uncertainty descriptions into the design process. In $\mathcal{H}_\infty$ control, the objective is to minimize the worst-case gain from disturbances to regulated outputs, yielding controllers that maintain performance under bounded perturbations. Extensions such as μ–synthesis address structured uncertainties through iterative D–K optimization, enabling more refined robustness guarantees in MIMO settings (Shao et al., 2025). Robust control also incorporates enhanced analysis and design tools based on Integral Quadratic Constraints (Michalowsky et al., 2021), which extend robustness guarantees to nonlinear and interconnected systems.

This type of control is widely applied in safety-critical and uncertainty-dominated environments, including aerospace systems, autonomous vehicles, and power and energy networks, where maintaining performance under parameter variations and external disturbances is essential. While robust control focuses on handling uncertainty, MPC offers a complementary framework based on constraint-aware optimization.

4. Advanced techniques in nonlinear observer design

The design of observers for nonlinear systems has long been a challenging and critical area in control theory. Unlike linear observers, nonlinear observers must contend with the complexities of strong nonlinearities, dynamic uncertainties, and the need for global state estimation under varying operating conditions. Nonlinear observers remain central to the AI-control interface because real-time decision-making under partial observability still hinges on accurate and computationally efficient state estimation (Bernard et al., 2022). Learning-augmented estimators and RL policies benefit from observer structures that inject physics-informed priors, stabilize training, and reduce data requirements, while accommodating model uncertainty and disturbances. Polynomial approximation bases, such as Laguerre functions, provide compact representations that can be embedded within probabilistic filters or learned policies, enabling tractable updates that satisfy latency and safety constraints in cyber–physical deployments (Asma & Kamel, 2009). This section reviews advanced nonlinear observer design techniques, focusing on recent research employing Laguerre polynomials to overcome these challenges while providing robust, accurate, and computationally efficient solutions. In addition to their analytical tractability, Laguerre expansions can be embedded within learning-based observer architectures, serving as physics-informed priors that stabilize the training and reduce the data requirements. When coupled with RL controllers, they further decrease sample complexity while preserving robustness to modeling errors, underscoring their role in AI-enhanced estimation.

4.1. Introduction to nonlinear observer design

Nonlinear observer design addresses the critical challenge of estimating the states of systems characterized by nonlinear dynamics. Unlike linear systems, where well-established frameworks, such as Luenberger observers, provide straightforward solutions, nonlinear systems often present complexities, such as nonlinearity in the state-space equations, unmodeled dynamics, and sensitivity to initial conditions. These complexities necessitate advanced methodologies capable of accommodating global behaviors and dynamic interactions within the system (Fliess & Sira-Ramirez, 2004; Wang & Gao, 2003).

Laguerre polynomials have emerged as a robust mathematical tool for nonlinear observer design, offering a globally valid basis for the approximation of nonlinear functions. Traditional approaches to nonlinear state estimation often rely on local linearization techniques such as those employed in the EKF (Simon, 2010). Although effective in mildly nonlinear systems, these methods struggle in highly nonlinear contexts because of their dependence on linear approximations around a specific operating point. Laguerre-based methods, by contrast, construct observers that approximate nonlinear functions over a broader range of operating conditions, ensuring greater accuracy and robustness (Garna et al., 2013).

The orthogonal nature of Laguerre polynomials with respect to the weight function e^{-t} is a key feature that facilitates an accurate representation of nonlinearities. By expanding the nonlinear functions

governing the system dynamics into a series of Laguerre polynomials, the essential characteristics of these functions can be efficiently captured without introducing significant computational overhead. This approach mitigates issues such as parameter identifiability and convergence delays, which are common in traditional nonlinear observer designs. Furthermore, the recursive nature of the Laguerre polynomials simplifies their computational implementation, making them practical for real-time applications (Han & Lam, 2015; Lerman, 1992).

In systems where disturbances, uncertainties, or model inaccuracies are prevalent, Laguerre-based nonlinear observers provide an efficient global approximation method that reduces the estimation errors compared with local linearization techniques. Their robustness is particularly beneficial for nonlinear and moderate-dimensional systems (Geng & Marmarelis, 2016). However, for high-dimensional cases, additional complexity reduction techniques, such as sparse representation or hybrid adaptive filtering, may be required to maintain the computational efficiency. Moreover, when operating in environments dominated by stochastic disturbances and measurement noise, probabilistic estimation mechanisms are necessary to complement the deterministic advantages of Laguerre-based designs (Liu et al., 2018).

Although Laguerre-based observers offer efficient nonlinear approximations and computational simplicity, they are not inherently equipped to handle the stochastic uncertainties present in real-world environments. Hybrid observer designs have been proposed that integrate Laguerre expansions within probabilistic estimation frameworks to enhance the robustness. In such approaches, the system dynamics approximated using a Laguerre basis are embedded into the stochastic filter structures. For instance, an EKF can utilize Laguerre-based approximations (Kumar et al., 2021) during the state prediction step, with the update step adjusting both the state estimate and covariance matrix based on noisy measurements (Lopez & Cid, 2020; Zhou & Deng, 2017). Alternatively, particle filters have been applied to sample trajectories governed by Laguerre-approximated dynamics, offering flexibility in handling nonlinear and non-Gaussian noise distributions (Chen & Shi, 2010). These hybrid designs enhance the estimation resilience against process disturbances and measurement noise, thereby expanding the applicability of Laguerre observers to complex, uncertain, and dynamic environments (Arasaratnam & Haykin, 2009; Pinheiro & Silveira, 2023; Schuck et al., 2022).

Nonetheless, they may perform poorly in real-time adaptive systems, such as autonomous vehicles on extreme terrain, power grids during sudden load fluctuations, or hypersonic flight control, where high-frequency aerothermal effects introduce complexities that exceed the capabilities of Laguerre-based approximations. In such cases, adaptive AI-driven techniques, including RL- and DL-based estimators, can enhance performance by dynamically adjusting to system variation. Model-free methods are also gaining traction in large-scale control applications, where predefined polynomial approximations may be insufficient (Kiumarsi et al., 2017).

The following formulations and comparative insights synthesize the key concepts from existing studies and lay the foundation for the mathematical treatment of advanced nonlinear observer designs.

4.2. Mathematical foundations

Advanced nonlinear observer designs often rely on systematic mathematical formulations to achieve robust and efficient state estimation in the presence of nonlinearities, uncertainties, and measurement noises. In this subsection, we outline the key structures, including polynomial expansions that underpin contemporary observer architectures.

More specifically, the mathematical framework for nonlinear observer design is based on the principle of approximating nonlinear system dynamics using orthogonal basis functions with Laguerre polynomials (Verhulst, 2012; Walker, 2013). Consider a nonlinear dynamical system described by the equations

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)), \quad \mathbf{y}(t) = h(\mathbf{x}(t), \mathbf{u}(t)), \tag{19}$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ represents the system state, $\mathbf{u}(t) \in \mathbb{R}^m$ is the control input, and $\mathbf{y}(t) \in \mathbb{R}^p$ is the measurable output. The functions $f(\mathbf{x}, \mathbf{u})$ and $h(\mathbf{x}, \mathbf{u})$ capture the nonlinear dynamics and measurement processes, respectively. The objective is to design an observer that estimates $\mathbf{x}(t)$ based on $\mathbf{y}(t)$ and $\mathbf{u}(t)$ (Khalil & Grizzle, 2002).

Laguerre polynomials, defined by the recurrence relation

$$L_{k+1}(t) = \frac{(2k+1-t)L_k(t) - kL_{k-1}(t)}{(k+1)}, \quad L_0(t) = 1,$$
$$L_1(t) = 1 - t \tag{20}$$

form the basis for approximating the nonlinear functions $f(\mathbf{x}, \mathbf{u})$ and $h(\mathbf{x}, \mathbf{u})$. These functions are expressed as finite series expansions

$$f(\mathbf{x}, \mathbf{u}) \approx \sum_{k=0}^{N} a_k(\mathbf{x}, \mathbf{u}) L_k(t), \quad h(\mathbf{x}, \mathbf{u}) \approx \sum_{k=0}^{N} b_k(\mathbf{x}, \mathbf{u}) L_k(t). \tag{21}$$

The coefficients $a_k(\mathbf{x}, \mathbf{u})$ and $b_k(\mathbf{x}, \mathbf{u})$ represent the projections of the system's nonlinear dynamics and input contributions onto the Laguerre basis functions, respectively, enabling the efficient modeling of complex behaviors in the observer update. In addition, $a_k(\mathbf{x}, \mathbf{u})$ and $b_k(\mathbf{x}, \mathbf{u})$ are determined using system identification and optimization techniques. The orthogonality of the Laguerre polynomials ensures that the coefficients are unique and computationally efficient to compute, reducing numerical instability in the observer design (Van den Hof et al., 2000).

Laguerre polynomials provide a basis for approximating nonlinear system dynamics over a defined interval. The summation in Eq. (19) represents the expansion of nonlinear functions in terms of Laguerre basis functions, allowing for the compact modeling of complex dynamics across a broad operating range while maintaining high approximation accuracy. This approximation is then incorporated into the observer structure through the recursive form in Eq. (20), offering a numerically efficient implementation for real-time state estimation. The orthogonality property of the Laguerre basis minimizes the cross-correlation between the basis components, thereby simplifying the estimation process and improving the numerical stability in observer computations (Billings, 2013; Wahlberg, 2002).

Hence, the observer is constructed as

$$\dot{\hat{\mathbf{x}}}(t) = f(\hat{\mathbf{x}}(t), \mathbf{u}(t)) + K(\mathbf{y}(t) - h(\hat{\mathbf{x}}(t), \mathbf{u}(t))), \tag{22}$$

where $\hat{\mathbf{x}}(t)$ is the estimated state vector, and K is the observer gain matrix. Substituting the Laguerre polynomial expansions for $f(\mathbf{x}, \mathbf{u})$ and $h(\mathbf{x}, \mathbf{u})$ yields a parameterized observer equation

$$\dot{\hat{\mathbf{x}}}(t) = \sum_{k=0}^{N} a_k(\hat{\mathbf{x}}(t), \mathbf{u}(t)) L_k(t) + K\left(\mathbf{y}(t) - \sum_{k=0}^{N} b_k(\hat{\mathbf{x}}(t), \mathbf{u}(t)) L_k(t)\right). \tag{23}$$

The observer gain K is typically designed to ensure the convergence of $\hat{\mathbf{x}}(t)$ to $\mathbf{x}(t)$. This involves analyzing the stability of the error dynamics

$$\mathbf{e}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t), \tag{24}$$

which is governed by

$$\dot{\mathbf{e}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)) - f(\hat{\mathbf{x}}(t), \mathbf{u}(t)) - K(h(\mathbf{x}(t), \mathbf{u}(t)) - h(\hat{\mathbf{x}}(t), \mathbf{u}(t))). \tag{25}$$

Stability analysis, often using Lyapunov functions, ensures that the error $\mathbf{e}(t)$ converges asymptotically to zero. The use of Laguerre polynomials to approximate $f(\mathbf{x}, \mathbf{u})$ and $h(\mathbf{x}, \mathbf{u})$ significantly simplifies the derivation of Lyapunov functions, as the orthogonality property facilitates the decomposition of error terms into independent components (Bouzaouache & Braiek, 2008; Lakshmikantham et al., 2013).

Laguerre-based nonlinear observers exhibit superior performance in handling high-dimensional nonlinear systems, in which conventional methods often fail to ensure convergence and robustness. By leveraging the mathematical properties of orthogonal functions, these observers provide a versatile and reliable framework for real-time state estimation in dynamic and uncertain environments (Kang et al., 2013; Saghatoleslami & Khaksar Toroghi, 2011).

4.3. Observers for descriptor and DAE systems

Descriptor or DAE systems introduce algebraic constraints in addition to differential dynamics, which complicates the observer design process. Unlike in ODE models, the estimated trajectories must not only track the system dynamics but also remain consistent with the algebraic invariants imposed by the descriptor structure (Rabier & Rheinboldt, 2002). As a result, classical observer formulations are inadequate for ensuring both convergence and constraint satisfaction in such systems.

Therefore, specialized observer approaches have been developed for DAE frameworks. In particular, projection-based and generalized Riccati formulations have been proposed to estimate the dynamic states while ensuring algebraic consistency (Tripathi et al., 2024). Extensions of Kalman and Luenberger observers incorporate index-reduction or projection techniques to maintain consistency with the algebraic manifold at each iteration (Mehrmann & Stykel, 2006). Optimization-based observers represent another direction in which state trajectories are obtained by solving constrained estimation problems that jointly satisfy the differential and algebraic equations. More recently, hybrid and data-driven extensions have coupled DAE models with neural surrogates or RL policies to approximate unmodeled dynamics while preserving constraint satisfaction (Rao et al., 2003).

These developments are particularly relevant in domains such as power grids, multi-body mechanical assemblies, and chemical process control, where algebraic constraints (e.g., Kirchhoff's laws or conservation relations) cannot be ignored (Pai, 1989). Embedding the descriptor-system structure into observer design provides a natural bridge between classical estimation theory and emerging AI-enhanced frameworks, ensuring both physical consistency and adaptability to uncertainty (Campbell & Gear, 1995).

Beyond descriptor-oriented frameworks, Laguerre basis expansions within the Volterra framework have been introduced for nonlinear processes and systems characterized by significant disturbances and time delays (e.g., reactors). Moreover, these compact representations enable offset elimination and improved disturbance rejection; however, their validation has so far remained limited to simulation studies (Zhang et al., 2010).

Overall, DAE-oriented and Laguerre–Volterra observer formulations extend the applicability of estimation theory to constrained and time-delayed environments, forming a conceptual link between physically consistent modeling and learning-based adaptive estimation schemes.

4.4. Comparative evaluation of Laguerre-based and conventional observers

Nonlinear observer design using Laguerre polynomials offers significant advantages over traditional methods such as EKF, HGO, and Lyapunov-based observers. Although these conventional techniques have been extensively used for state estimation, they encounter critical limitations when addressing highly nonlinear dynamics, significant uncertainties, and global approximation challenges.

HGOs are widely regarded for their rapid convergence properties, which are achieved through HGO settings. However, this approach is sensitive to noise and model uncertainties because excessively high gains amplify the measurement noise, resulting in degraded performance or instability (Tornambè, 1992). While such foundational analyses established the theoretical ability of HGOs to achieve arbitrarily fast convergence, later studies emphasized that they are also prone to peaking effects and strong noise sensitivity, which restrict their practical use in noisy environments (Khalil & Praly, 2014).

An early contribution was the real-time implementation of high-gain and Kalman-type observers on a direct current (DC) motor platform using a real-time application interface (RTAI)/Scilab. Three variants were compared: a classic high-gain Luenberger observer, a high-gain EKF, and an adaptive gain EKF. The adaptive scheme adjusts the high-gain parameter online using an innovation measure that interpolates between the EKF-like noise rejection and high-gain convergence. Experiments confirmed the feasibility of hard real-time systems, demonstrating that adaptive gain filtering can achieve smoother convergence than fixed HGO while retaining robustness against perturbations (Boizot et al., 2007).

Additionally, HGOs were integrated into a Lyapunov-based switching control framework to address parametric uncertainties in nonlinear systems. In this approach, candidate controllers are paired with Lyapunov functions, whereas the HGO supplies the necessary state and Lyapunov derivative estimates. A supervisory logic enforces Lyapunov decrease conditions to select the appropriate controller, ensuring stability without restrictive assumptions, such as minimum-phase dynamics or a known input gain sign. This design mitigates the peaking effects associated with static high-gain configurations, with simulation results confirming improved transient performance compared to the conventional HGO-based feedback (Freidovich & Khalil, 2007).

A robust extension of the EKF (REKF) is proposed for nonlinear systems with stochastic uncertainties. The formulation modifies the Riccati updates to enforce an optimized bound on the estimation error covariance, explicitly incorporating model uncertainties and linearization errors. Compared with the standard EKF and H_∞ filters, this design achieves greater robustness against process and measurement noise. Numerical studies on X-ray pulsar positioning confirmed improved estimation consistency, although validation remained limited to simulations rather than hardware tests (Kai et al., 2009).

Hybrid nonlinear observers have been proposed that integrate the Extended HGO (EHGO) with EKFs. In this structure, the EHGO estimates the output derivatives and provides a virtual output, which the EKF exploits to estimate internal dynamics. The resulting design ensures local convergence in general nonlinear cases and semi-global convergence when the internal dynamics are nonlinear. Simulation studies on a synchronous generator and the TORA benchmark demonstrated the effectiveness of the proposed method in scenarios where standalone HGOs or EKFs failed to deliver reliable estimations (Boker & Khalil, 2013).

A Laguerre–Wavelet Wiener model for nonlinear system identification was proposed in Aadaleesan et al. (2008), where Laguerre filters capture global system dynamics and wavelet networks model localized nonlinearities. This hybrid representation achieves higher accuracy and parsimony than polynomial Laguerre models, as validated by simulations of bioreactor and laboratory pasteurization processes. Although this approach improves robustness under noisy conditions, the reported applications remain confined to simulations and laboratory-scale experiments only.

Another line of research integrates Laguerre expansions with neural estimators, such as the Laguerre-extreme learning machine (ELM) Wiener model. In this architecture, the Laguerre filters compactly represent the system dynamics, whereas the ELM provides a fast, nonlinear approximation with minimal parameters. This framework achieves a low mean squared error (MSE) in single-input, single-output (SISO) nonlinear system identification and extends naturally to generalized predictive control, where it offers a lightweight alternative to more complex recurrent neural estimators. In this study, the suggested model was benchmarked against several neural estimators, including wavelet recurrent NN (WRNN), Chebyshev RNN (CReNN), dynamic feedforward NNs (DFNN), and conventional multilayer NN (MNN) (Tang et al., 2016).

A nonlinear disturbance observer for serial n-link manipulators was developed using Lyapunov stability theory, which demonstrated how Lyapunov-based designs can ensure global asymptotic convergence. This approach requires tuning of only a single parameter and avoids explicit disturbance modeling, highlighting a simpler alternative to traditional nonlinear observers. Simulation studies on three-link manipulators confirmed the effective rejection of both internal uncertainties, such as unknown payloads, and external disturbances, such as joint

friction, while maintaining accurate trajectory tracking (Nikoobin & Haghighi, 2009).

A nonlinear extended-state ARX-Laguerre PI observer (EALPIO) was introduced for fault diagnosis in induction motor bearings. The approach combines ARX-Laguerre modeling with a proportional–integral observer and sliding mode residual generator, yielding robust residuals in the presence of uncertainties and noise. Experiments across variable speeds and fault sizes showed accurate detection and isolation of multiple bearing faults, outperforming conventional ARX–Laguerre PI observers (ALPIO) in terms of diagnostic accuracy and robustness (Piltan & Kim, 2019).

Traditional observer designs often suffer from a rapid increase in complexity as the system order increases, thereby limiting their applicability to large-scale nonlinear processes. To address this, Laguerre–Volterra observers embed orthonormal Laguerre bases into Volterra kernel expansions to produce compact observer-controller structures with reduced parameterization requirements. This formulation has been applied to chemical reactor systems, demonstrating offset elimination, improved disturbance rejection, and provable stability and tracking performance, thereby offering a scalable alternative to conventional PID-based schemes and NMPC (Zhang et al., 2008).

Another approach introduces Laguerre multiple-model observers, in which nonlinear dynamics are captured using a decoupled Laguerre expansion. This structure significantly reduces the parameter estimation burden compared to conventional nonlinear observers, such as Nonlinear AutoRegressive Moving Average with eXogenous inputs (NARMAX) models, while retaining the modeling accuracy. Simulation studies confirmed stable convergence, highlighting the potential of this framework for scalable and computationally efficient observer design in embedded and online identification contexts (Marouani et al., 2015).

Another study extended the Laguerre-based observer design to systems with time delays. In this approach, the delay dynamics are represented by a coupled partial differential equation (PDE) and an ODE formulation. The prescribed time convergence of all state estimates is enforced by constructing time-varying gains using Laguerre polynomials, symmetric polynomials and Vandermonde matrices. Compared with traditional delay-compensated observers, this design achieves smooth finite-time convergence of both the states and their derivatives. However, its reliance on rapidly varying gains increases the sensitivity to measurement noise, and its validation has so far been restricted to simulation studies (Espitia et al., 2022).

An alternative formulation employs Laguerre polynomials for nonlinear observer design in the canonical observability form, addressing state estimation when the system dynamics are only partially known. By approximating the nonlinear aggregation function using a linear combination of Laguerre polynomials, the framework reduces the estimation problem to parameter identification within an orthogonal basis set. Validated on a bioreactor model, the proposed observer demonstrated stable convergence and low estimation errors for both measured and unmeasured states, confirming the suitability of the Laguerre-based expansions for robust nonlinear identification under limited prior knowledge (Trigka & Dritsas, 2022).

A recent study in automotive engineering demonstrated the use of Laguerre compensators in steer-by-wire (SBW) systems, where digital compensators are computed via pole placement using Schur polynomials derived from Laguerre bases. This approach ensured closed-loop stability for strictly proper plants and improved the control accuracy compared with PID designs. The performance was assessed using the error-to-signal ratio (ESR), which quantifies the tracking error relative to the reference signal, and the adjusted ESR (AESR), which refines this metric by compensating for actuation delays. Experimental validation on embedded vehicle control units (VCU) with controller area network (CAN) protocols confirmed significant reductions in both ESR and AESR, highlighting the practicality of this approach in safety-critical automotive environments (Arceo et al., 2024).

In summary, Laguerre-based observers represent a robust and scalable alternative to traditional methods, effectively addressing the challenges related to strong nonlinearity, noise sensitivity, and computational inefficiency. Table 6 provides a qualitative comparative analysis, focusing on the structural characteristics, theoretical attributes, and general performance advantages of the different observer designs. Beyond these conventional dimensions, the table also considers AI/learning compatibility and applicability to descriptor systems, reflecting recent trends in integrating observers with RL and extending the estimation to differential–algebraic formulations. This conceptual evaluation highlights the enhanced modeling flexibility and robustness offered by polynomial-based approaches, such as the Laguerre expansion, particularly in data-driven and large-scale CPS contexts.

In contrast, Table 7 presents a quantitative performance comparison across diverse application domains, detailing metrics such as estimation accuracy, computational efficiency, baseline references, and real-time implementation capabilities. The summarized results, drawn from both simulation-based studies and experimental validations, emphasize the practical effectiveness of Laguerre-based observers in balancing modeling precision and computational tractability, particularly under real-time constraints. Together, these comparative analyses demonstrate that Laguerre-based observers constitute a valuable and promising toolset within modern control theory, capable of delivering robust state estimation and accurate nonlinear system modeling in complex operating environments. Nevertheless, current empirical reports remain largely confined to conventional nonlinear settings, with limited exploration of descriptor-system observers or AI-enhanced learning architectures, highlighting promising directions for future studies. The comparative advantages outlined here motivate their application across various domains, including energy, transportation, and biomedical systems, which are discussed in the next section.

5. Applications of control theory

Control theory provides a mathematical foundation for regulating and stabilizing complex engineered systems across diverse domains. Its significance extends beyond traditional dynamical models to encompass large-scale, networked infrastructures, where descriptor system formulations and advanced observer architectures are indispensable for addressing algebraic constraints, uncertainties, and partial observability. The applications in this section (Fig. 9) are presented with a domain-specific emphasis on control-theoretic formulations and methodologies, highlighting how differential–algebraic models, nonlinear observers, and predictive strategies enable robust, real-time operation.

5.1. Energy and power systems

Energy systems, particularly electrical power grids, represent archetypal large-scale infrastructures in which control theory underpins both stability and operational efficiency. The governing equations of such networks naturally take the form of DAE, with algebraic constraints arising from Kirchhoff's current and voltage laws and differential components describing generator and dynamic load behavior (Gu & Green, 2022; Machowski et al., 2020). These descriptor models necessitate observer designs capable of handling hidden states and unmeasured disturbances while respecting the algebraic coupling inherent in the network (Grigsby, 2007).

MPC has become a dominant methodology in frequency and voltage regulation, leveraging finite-horizon optimization to accommodate nonlinearities, operating constraints, and fluctuating renewable inputs (Mayne et al., 2000). In parallel, LQR and Gaussian extensions remain central to balancing performance and robustness, particularly for secondary control in large-scale interconnected grids (Vittal et al., 2019). To ensure reliability under uncertain demand and stochastic disturbances, adaptive and robust observers are integrated into state

Table 6
Qualitative comparative analysis of state estimation methods, extended with AI compatibility and descriptor-system applicability.

Observer type	Handling nonlinearities	Noise robustness	Computational efficiency	Global approximation capability	AI/Learning compatibility	Applicability to DAE systems
EKF	Weak (local linearization)	Moderate	Moderate	Limited	Low	Limited (index reduction only)
HGO	Moderate	Low (noise sensitive)	High	Limited	Low–Moderate	Low
Lyapunov-based	Strong (specific systems)	Moderate	Low (high complexity)	Moderate	Moderate	Moderate (with complexity)
Laguerre-based	Strong (global approximation)	High (robust to noise)	High (efficient scalability)	Strong	High	High

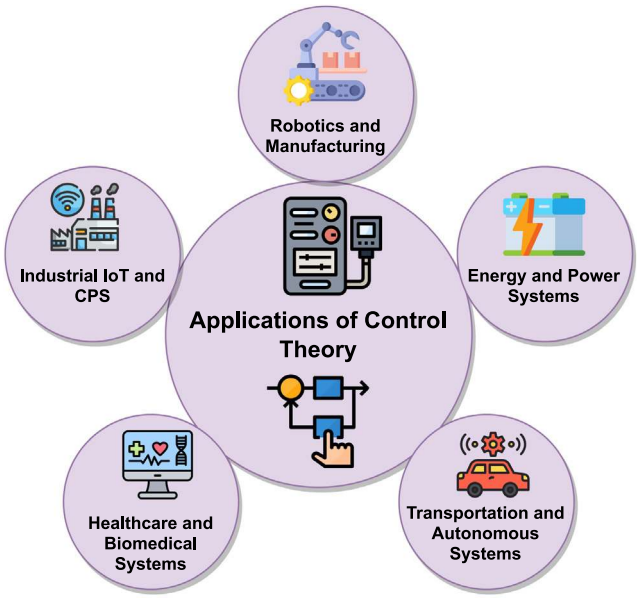


Fig. 9. Representative Application Domains of Modern Control Theory.

estimation pipelines to provide enhanced fault detection and resilience against sensor failures. These observer-based schemes extend beyond classical static estimators, enabling the dynamic monitoring of voltage profiles, phase angles, and distributed energy resources under real-time latency constraints (Abur & Exposito, 2004).

The synergy of descriptor system modeling with predictive and observer-based control illustrates how fundamental control-theoretic tools ensure frequency stability, mitigate cascading failures, and support the secure integration of renewable generation (Chiang, 2011). Beyond physical reliability, recent research emphasizes resilience against cyber–physical disruptions, where descriptor-based observers and fault-tolerant controllers are adapted to detect stealthy attacks or sensor manipulations while maintaining stability under Kirchhoff-constrained dynamics. By grounding system operation in rigorous mathematical structures, control theory continues to serve as the backbone of energy and power system resilience (Teixeira et al., 2010).

5.2. Transportation and autonomous systems

Transportation and aerospace platforms are canonical domains in which control theory directly affects safety, performance, and efficiency. In automotive systems, control design underlies safety-critical modules such as the SBW, adaptive cruise control, and lane-keeping (Isermann, 2022). These systems are often modeled with nonlinear dynamics, where actuator saturation, tire–road interactions, and time-varying delays complicate state estimation. To address these challenges, nonlinear observers, including sliding-mode and adaptive structures, have been employed to reconstruct unmeasured states, such as lateral velocity and tire forces, with robustness against disturbances (Veluvolu & Soh, 2011). Such observer-enhanced controllers enable fault-tolerant performance by isolating sensor faults and ensuring continued operation under degraded conditions (Yang et al., 2015).

Aerospace applications further demonstrate the necessity for advanced control-theoretic constructs. For flight control, Lyapunov-based disturbance observers and HGO are used to compensate for aerodynamic parameter uncertainties and rapidly varying external loads, ensuring stable attitude tracking under turbulence and actuator delays (Yang et al., 2013). These methods are complemented by gain-scheduled MPCs that integrate observer feedback to handle constraints on flight envelopes and actuator saturation (Hu et al., 2014). In spacecraft guidance and reentry vehicles, descriptor-system formulations capture coupled translational and rotational dynamics, with observer-based designs enabling state reconstruction when sensor measurements are limited or delayed by communication lags (Shtessel et al., 2014).

In both automotive and aerospace contexts, the integration of observer-based designs with classical control frameworks, such as LQR/LQG and MPC, ensures that nonlinear dynamics, delays, and uncertainties are systematically accounted for. The resulting architectures deliver provable robustness guarantees while retaining computational tractability, thereby meeting the stringent real-time requirements of safety-critical transportation systems (Di Cairano & Bemporad, 2009; Shearer & Cesnik, 2008).

5.3. Healthcare and biomedical systems

The biomedical applications of control theory illustrate how rigorous mathematical frameworks can directly influence patient safety and treatment efficacy. Closed-loop drug delivery systems, such as insulin infusion for diabetes management or anesthesia regulation during surgery, require controllers capable of maintaining stability despite patient-specific nonlinearities, unmodeled dynamics, and delays in measurement. Adaptive and robust control laws have been particularly effective in this setting, ensuring bounded responses while maintaining performance across wide inter- and intra-patient variabilities (Hovorka, 2011; Liu et al., 2006).

The design of observer-based strategies is equally critical because many physiological variables, including plasma insulin concentration and neural activity, cannot be measured directly in real time (Homayounzade, 2022). Delay-compensating observers and robust nonlinear estimators are employed to reconstruct these hidden states from sparse and noisy sensor signals, thereby enabling safe and responsive actuation (Sakthivel et al., 2018). Lyapunov-based observer designs provide explicit stability guarantees, ensuring that reconstructed physiological states remain reliable even under abrupt changes in patient conditions or sensor dropouts (Alessandri, 2020).

In addition to drug delivery, control theory supports advanced biomedical devices, such as neural stimulation systems, cardiac assist devices, and brain–computer interfaces. These systems operate under stringent safety constraints, where actuation must respect both physiological and regulatory limits (Esposito et al., 2021). The MPC has

Table 7
Comparative real-world performance of Laguerre-based and traditional observers.

Method	Application domain	Estimation performance	Efficiency & Complexity	Comparison baseline	Real-time applicability
HGO (Khalil & Praly, 2014)	Nonlinear feedback; robust control of minimum-phase systems	Fast/global convergence; limited by noise amplification and peaking	High (tunable via ϵ), but sensitive to noise–robustness trade-off	Classical output feedback; full state-feedback	Yes – theory and simulations; constrained in noisy settings
Adaptive-Gain EKF (Boizot et al., 2007)	DC motor (hard real-time platform)	Faster convergence from large errors; smoother tracking under perturbations	Hard real-time feasible (RTAI/Scilab)	high-gain EKF, HGO (Launberger), adaptive gain EKF	Yes – validated in hard real-time experiments
Switching HGO (Freidovich & Khalil, 2007)	Parametric uncertain nonlinear systems	Improved transient performance; avoids peaking	Moderate (supervisory switching)	Classical HGO-based feedback	Partial – simulation validation only
Robust EKF (Kai et al., 2009)	Pulsar positioning system (nonlinear, stochastic uncertainties)	Improved estimation consistency; reduced sensitivity to uncertainty	Modified Riccati updates ensuring bounded error covariance	EKF, H_∞ filter	Partial – simulation validation only
Hybrid EHGO + EKF (Boker & Khalil, 2013)	Power systems (synchronous generator), TORA	Local convergence (general); semi-global when internal dynamics are linear; handles unstable zero dynamics	Higher complexity (EHGO derivative estimation + EKF Riccati)	Standalone EKF or HGO	Partial – simulation only
Laguerre-Wavelet Wiener Observer (Aadaleesan et al., 2008)	Bioreactor and pasteurization processes	Improved accuracy and robustness under noisy data	Hybrid Laguerre-wavelet representation; compact basis	Polynomial Laguerre and classical neural models	Partial – simulation and lab-scale validation only
Laguerre-ELM Wiener Model (Tang et al., 2016)	SISO nonlinear system identification and predictive control	Low MSE; accurate nonlinear modeling with few parameters	Fast learning with minimal parameterization	WRNN, CReNN, DFNN, MNN	Yes – lightweight architecture suitable for online use; validated in simulations & benchmarks
Lyapunov-based nonlinear disturbance observer (Nikoobin & Haghighi, 2009)	N-link robotic manipulators	Simulation-based precise disturbance estimation	Moderate cost due to Lyapunov analysis	Sliding Mode Observer	Partial – Simulation-based validation; not tested on physical hardware
EALPIO (Piltan & Kim, 2019)	Fault diagnosis in induction-motor bearings	High fault detection accuracy under varying operating conditions	Robust under motor speed variation; sliding-mode residuals	ALPIO	Yes – real-time evaluation under variable speed
Laguerre–Volterra Observer (Zhang et al., 2008)	Chemical reactor system	Stable tracking and disturbance rejection with offset elimination	Compact Volterra expansion with Laguerre basis (reduced parameterization)	Classical PID/NMPC	Partial – simulation and lab-scale validation; no industrial testing
Laguerre Multiple-Model Observer (Marouani et al., 2015)	Nonlinear MIMO systems	Stable convergence validated in simulations; reduced parameter estimation burden	Decoupled Laguerre model structure; computationally efficient	Conventional nonlinear observer models (e.g., NARMAX)	Yes – reduced structure suitable for embedded and real-time implementation
Laguerre-based Prescribed-Time Observer (Espitia et al., 2022)	LTI systems with sensor delay	Smooth finite-time convergence of states and derivatives	Time-varying gains constructed via Laguerre polynomials, symmetric polynomials, and Vandermonde matrices	Classical delay-compensated observers	"Partial – simulation validation only; sensitive to measurement noise
Laguerre Observer (Trigka & Dritsas, 2022)	Biological system model	Low estimation error for both states	Fast convergence for specific θ	Not compared directly	Partial – Simulation-based validation; not tested on physical hardware
Laguerre Compensator (Arceo et al., 2024)	SBW automotive system	Reduced error metrics (ESR, AESR) (ESR, AESR)	Real-time embedded VCU with CAN protocol	Classical PID	Yes – Real-time validated on SBW platform

been deployed to enforce such constraints while maintaining therapeutic effectiveness, often with observers serving as the interface between noisy biosignals and control input computation. In rehabilitation robotics and prosthetics, observer-augmented controllers stabilize nonlinear musculoskeletal dynamics and compensate for uncertainties in human–machine interactions (Aljuboury et al., 2022; Mashud et al., 2025).

These examples demonstrate the ability of control theory to extend beyond industrial settings into domains where human safety is paramount. By embedding observer-based estimation into robust and

adaptive control architectures, biomedical applications achieve both reliability and personalization, two attributes that are otherwise difficult to reconcile in complex, time-varying physiological systems (Bequette, 2012).

5.4. Industrial IoT and CPS

Modern industrial environments are increasingly structured as large-scale CPS, where physical processes interact tightly with distributed computation and communication layers (Rajkumar et al., 2010). Control theory provides the backbone for ensuring stability, safety, and performance under interconnected and delay-prone conditions (Hespanha et al., 2007). A key modeling paradigm is the use of hybrid and descriptor-type formulations, where discrete event dynamics are coupled with continuous-time DAE. This representation captures both the physical plant constraints and the logical interactions dictated by digital controllers and communication protocols (Antsaklis, 2000).

Within industrial IoT settings, distributed control strategies are essential for managing geographically dispersed plants and sensor networks. Consensus-based controllers and distributed MPC architectures allow coordination among multiple agents while respecting communication bandwidth limits (Maestre et al., 2014). Observers play a central role in maintaining reliability, particularly in the presence of packet drops, latency, or adversarial disturbances in the network (Zhang et al., 2001). Robust and resilient observer designs ensure that the estimation accuracy is preserved, thereby enabling stable closed-loop operation even under partial or corrupted measurements (Ding, 2008).

Industrial CPS also faces stringent safety and security requirements. Control-based anomaly detection mechanisms leverage residuals from observer structures to identify faults, cyberattacks, or equipment degradation in real time (Pasqualetti et al., 2013). Sliding-mode and adaptive observer designs provide robustness against both unstructured process disturbances and malicious perturbations injected through compromised sensors or actuators (Edwards et al., 2000). By embedding such resilient estimation layers, industrial automation systems can maintain stability and performance guarantees despite the growing vulnerabilities in networked infrastructures (Teixeira et al., 2015).

These methods highlight the fact that control theory remains indispensable for industrial IoT and CPS, where the challenges of distributed coordination, hybrid modeling, and cyber–physical security converge. By integrating observers with advanced feedback strategies, modern industrial systems achieve reliable operation at scale while respecting the latency and safety constraints imposed by heterogeneous plants and communication networks (Dörfler & Bullo, 2014).

5.5. Robotics and manufacturing

Robotics and manufacturing environments impose stringent demands on control systems, where precision, robustness, and adaptability must be achieved in real time. Robotic manipulators represent benchmark systems for nonlinear control because their dynamics are highly coupled, time-varying, and sensitive to model uncertainties. Classical approaches, such as Lyapunov-based adaptive control and sliding mode strategies, remain fundamental for ensuring trajectory tracking accuracy under parameter variations and external disturbances (Ciliz, 2005; Wei, 2018). These methods guarantee stability while handling payload changes, frictional effects, and actuator nonlinearities that are inherent to industrial robotics (Sun, 2022).

Model calibration and parameter identification are also essential for achieving precise and reliable robotic control. A related study on industrial robot calibration highlights advances in standardization, modeling accuracy, and performance evaluation (Li et al., 2021). Diversified regularization schemes have been developed to enhance manipulator calibration in nonlinear settings (Li et al., 2022). These approaches highlight the link between model accuracy and control performance in modern robotic systems.

The role of observers in robotic systems is equally critical. Fault detection observers are deployed to detect anomalies in joint actuators, encoders, or force sensors, enabling fault-tolerant control without interrupting the operation (Ma & Yang, 2016). Robust state observers also provide accurate estimates of unmeasured joint velocities and external disturbances, which are vital for achieving precise positioning and ensuring safety in collaborative robotic systems (Izadbakhsh & Nikdel, 2022). In high-precision applications, observer-enhanced feedback architectures reduce the latency in estimating fast dynamics, thereby enabling tighter control loops with minimal overshoot (Huo et al., 2025).

Manufacturing systems increasingly rely on predictive control frameworks to balance efficiency and robustness. MPC provides a natural framework for handling constraints in robotic assembly lines and collaborative multi-robot systems. MPC formulations explicitly account for latency and collision-avoidance constraints, allowing for the safe coordination of multiple manipulators operating in shared workspaces (Krämer et al., 2020). This is especially relevant in human–robot collaboration scenarios, where control architectures must reconcile safety requirements with real-time responsiveness (Elaamery et al., 2021; Oleinikov et al., 2024).

Beyond these classical and model-predictive formulations, bio-inspired neural control frameworks have also been explored for cooperative and domestic robotic systems. Such architectures integrate adaptive neural dynamics to achieve real-time coordinated control in uncertain environments, offering complementary perspectives to conventional model-based control (Khan, Li, & Cao, 2021).

Taken together, robotics and manufacturing applications exemplify how control theory enables robust, scalable, and safety-critical operations in highly dynamic environments. Through nonlinear feedback, resilient observers, and predictive strategies, robotic systems can achieve the precision and adaptability required in modern industrial ecosystems. Table 8 summarizes domain-specific applications of control theory by mapping the interplay between system models, control techniques, observer architectures, and representative use cases. The comparison highlights how different domains exploit distinct modeling formalisms, ranging from descriptor formulations in power grids to delay-differential structures in biomedical systems, and deploy tailored controllers, such as predictive, robust, or nonlinear schemes. Observers emerge as a unifying mechanism across all sectors, ensuring resilient state estimation and fault detection under real-time constraints. In contrast, the readiness column reflects the extent to which these approaches have transitioned from theory and simulation to hardware-in-the-loop (HIL), clinical validation, and embedded deployment.

The breadth of applications presented underscores the enduring relevance of control theory as a rigorous and adaptable framework for safety-critical, networked, and real-time systems. Nevertheless, as infrastructure expands in scale and interconnectivity, classical methods, while powerful, face intrinsic limits in addressing demands for adaptability, autonomy, and resilience under deep uncertainty. These limitations open the pathway for integration with emerging paradigms in AI, ML, quantum technologies, and sustainability-driven design, which will be the focus of the following section.

6. Integration with emerging technologies

The applications discussed in Section 5 demonstrate the foundational role of control theory in critical infrastructure. However, as these systems evolve into more interconnected, data-rich, and adaptive environments, classical methods face limitations in terms of scalability, resilience, and autonomy. Addressing these challenges requires the systematic integration of control-theoretic principles with emerging paradigms, such as ML, digital twins, cyber–physical architectures, quantum computing, and sustainability-driven design. This section examines these integrations, emphasizing how hybrid frameworks preserve the analytical rigor of control while expanding their reach towards adaptive and intelligent infrastructures.

Table 8
Domain-specific applications of control theory.

Domain	Core models	Main control techniques	Observer Use	Representative use cases	Real-Time/Deployment readiness
Energy and Power Systems (Abur & Exposito, 2004; Chiang, 2011; Grigsby, 2007; Gu & Green, 2022; Machowski et al., 2020; Mayne et al., 2000; Teixeira et al., 2010; Vittal et al., 2019)	DAE-based grid models (Kirchhoff constraints, stability DAEs)	PID for primary frequency control, MPC, LQR/LQG for voltage/frequency regulation, load balancing	Adaptive/robust observers for fault detection, state estimation	Frequency control in smart grids, microgrid stability, fault-tolerant operation, economic–energy co-optimization under uncertainty	Field/HIL
Transportation and Autonomous Systems (Di Cairano & Bemporad, 2009; Hu et al., 2014; Isermann, 2022; Shearer & Cesnik, 2008; Shtessel et al., 2014; Veluvolu & Soh, 2011; Yang et al., 2013, 2015)	Nonlinear vehicle/aerospace dynamics, hybrid models	N-PID and MPC for trajectory tracking, adaptive/sliding-mode control for fault tolerance, observer-based feedback for vehicle and UAV stabilization	Nonlinear/Lyapunov-based disturbance observers	SBW automotive, adaptive cruise, UAV trajectory stabilization, autonomous path planning under constraints	HIL/Flight tests
Healthcare and Biomedical Systems (Alessandri, 2020; Aljuboury et al., 2022; Bequette, 2012; Esposito et al., 2021; Homayounzade, 2022; Hovorka, 2011; Liu et al., 2006; Mashud et al., 2025; Sakthivel et al., 2018)	Physiological delay-differential models, pharmacokinetics	PID for infusion-rate control, constrained MPC for physiological safety limits, adaptive/robust control for patient-specific modeling	Delay-compensating and Lyapunov-based observers for safety-critical estimation	Closed-loop insulin delivery, anesthesia depth control, neural stimulation, autonomous drug titration systems	Clinical/Pre-clinical
Industrial IoT and CPS (Antsaklis, 2000; Ding, 2008; Dörfler & Bullo, 2014; Edwards et al., 2000; Hespanha et al., 2007; Maestre et al., 2014; Pasqualetti et al., 2013; Rajkumar et al., 2010; Teixeira et al., 2015; Zhang et al., 2001)	Hybrid discrete–continuous process models, distributed DAEs	Distributed PID and MPC for large-scale coordination, robust/resilient feedback for process automation, observer-based fault isolation under communication delays	Resilient/adaptive observers for anomaly detection and fault isolation	Large-scale process automation, chemical plants, water distribution networks, secure CPS operation under latency and attack scenarios	Testbeds/Pilots
Robotics and Manufacturing (Ciliz, 2005; Elaamery et al., 2021; Huo et al., 2025; Izadbakhsh & Nikdel, 2022; Khan, Li, & Cao, 2021; Krämer et al., 2020; Li et al., 2022, 2021; Ma & Yang, 2016; Oleinikov et al., 2024; Sun, 2022; Wei, 2018)	Nonlinear manipulator and kinematic models	Hybrid PID—MPC coordination for manipulators; Lyapunov-based adaptive and sliding-mode control; bio-inspired neural frameworks for collaborative autonomy; and data-driven calibration and parameter identification for improved modeling accuracy	Observers for fault detection, joint state estimation, and real-time calibration consistency	Collaborative robotics, precision manufacturing, human–robot interaction, adaptive assembly-line learning, and self-calibrating manipulators	Embedded real-time

6.1. AI and ML in control

The integration of ML into control theory has reshaped the modeling, observation, and regulation of dynamical systems. While traditional approaches rely on physics-based models and analytical designs, supervised and unsupervised learning have emerged as complementary tools for extracting system representations directly from operational data (Pillonetto et al., 2025). In supervised settings, regression and classification models facilitate system identification and predictive maintenance, whereas unsupervised clustering and dimensionality reduction methods support anomaly detection and reduced-order modeling in high-dimensional dynamical systems (Yu et al., 2025). These data-driven representations offer flexibility in domains where first-principles modeling is either infeasible or computationally prohibitive (Wu et al., 2025).

DL provides further expressivity by approximating highly nonlinear input–output mappings and embedding latent representations into the control pipelines. Neural network-based controllers have demonstrated strong performance in tasks requiring nonlinear compensation and adaptive behavior (Peng et al., 2022). In contrast, neural observers extend these capabilities by reconstructing unmeasured states from noisy and incomplete data (Zhu et al., 2023). When combined with Lyapunov-stable architectures or constrained optimization layers, deep models can be endowed with stability and safety guarantees that align with the rigorous requirements of safety-critical systems (Stiti et al., 2024).

The growing convergence of control theory with AI-driven optimization has led to architectures that combine learning, prediction, and decision-making within unified adaptive frameworks. Hybrid bio-inspired algorithms such as the Enhanced Beetle Antennae Search integrated with Zeroing Neural Network (BAS-ZNN) (Khan, Cao, Li, & Li, 2021) directly contribute to intelligent control by solving constrained and nonlinear optimization problems in real time. Related works extended these optimization-driven formulations beyond conventional physical systems, including neural-network-based fault prediction in energy infrastructures (Aghajari & AbdulRahim, 2024) and neural dynamic optimization for decision-making in economic settings (Cao et al., 2024). Together, these approaches highlight how neural and bio-inspired computation can enhance adaptability, prediction accuracy, and robustness across both classical control scenarios and emerging AI-enabled domains.

RL introduces a complementary paradigm in which adaptive controllers are trained through interaction with the environment rather than pre-collected datasets. Policy gradient and value-based methods have enabled real-time adaptation in uncertain environments, particularly when coupled with model-based RL that leverages approximate system dynamics (Greatwood & Richards, 2019). Hybrid approaches, where RL policies are regularized by physics-informed priors or embedded within predictive control frameworks, reduce sample complexity and ensure that learned strategies remain interpretable and deployable under resource and latency constraints (Joglekar et al., 2022).

Distributed control scenarios motivate the use of FL, where models are collaboratively trained across multiple agents without centralized data aggregation (Weber et al., 2024). In power grids, industrial IoT, and transportation networks, federated schemes allow controllers and observers to adapt to heterogeneous local conditions while preserving privacy and reducing communication overheads (Bhatti et al., 2025; Dritsas & Trigka, 2025a). This aligns with the growing demand for scalable, secure, and resilient control in networked infrastructure, where centralized learning approaches face bottlenecks in bandwidth, security, and fault tolerance (Dritsas & Trigka, 2025b).

Together, these AI-driven methodologies illustrate a spectrum of integration paths: supervised and unsupervised learning enhance modeling and anomaly detection, DL augments controllers and observers with expressive nonlinear representations, RL provides adaptive decision-making in uncertain environments, and FL enables collaborative intelligence in distributed systems. These paradigms do not replace classical control designs but extend them into domains where data availability, uncertainty, and scale challenge purely analytical approaches (Lawrence et al., 2024).

6.2. Digital twins and data-driven modeling

Digital twins represent a paradigm shift in control engineering, enabling high-fidelity virtual replicas of physical assets that evolve synchronously with their real-world counterparts (Kapteyn et al., 2021). Unlike conventional simulation models, digital twins integrate multiphysics descriptions, sensor data streams, and adaptive calibration pipelines, creating a dynamic surrogate capable of supporting control design, monitoring, and predictive maintenance. Within this framework, control theory serves as the unifying mathematical backbone, ensuring that digital twin models remain both dynamically consistent and computationally tractable under real-time constraints (Reich, 2025; Yao, Yang, Wang, & Zhang, 2023).

From a methodological perspective, digital twins bridge physics-based modeling with data-driven identification. Descriptor-system formulations, reduced-order modeling, and system identification techniques enable the construction of twin models that capture both the fundamental conservation laws and latent dynamics extracted from operational data (Kapteyn et al., 2022). Observer-based estimation plays a pivotal role in synchronizing the twin with the evolving physical plant, compensating for sensor noise, missing data, and time-varying disturbances (Peterson et al., 2025). By embedding adaptive and robust observers, digital twins achieve continuous state alignment, even when partial or corrupted measurements are fed from distributed sensors (Lei et al., 2023).

The integration of data-driven control architectures further extends the utility of digital twins. ML surrogates and operator-inference models provide real-time approximations of nonlinear dynamics, which, when combined with predictive control formulations, support the runtime optimization of system performance (Gratius et al., 2023). Importantly, these surrogates are not deployed as black boxes but as hybrid models that preserve their physical interpretability while leveraging their data-driven adaptability. This duality ensures that digital twins can function as safe and certifiable platforms for decision-making in safety-critical domains (Bárkányi et al., 2021).

Practical deployments of digital twin-enabled control span a wide range of sectors, from predictive maintenance in industrial automation to stability monitoring in energy grids and personalized medicine in biomedical applications. In each case, the twin serves not only as a monitoring layer but also as a closed-loop participant, enabling what-if analysis, fault anticipation, and resilience assessment under dynamic operating conditions. By coupling real-time sensing, hybrid modeling, and control-theoretic rigor, digital twins transform static system representations into interactive computational platforms that evolve alongside their physical counterparts, thereby redefining the design, validation, and operation of control systems in complex infrastructures (Mchirgui et al., 2024; Van Dinter et al., 2022).

6.3. CPS and IoT with emerging control

CPS and IoT increasingly provide the architectural substrate through which modern control systems are deployed, scaled, and adapted. Unlike domain-specific applications, where control is confined to a physical process, CPS and IoT platforms integrate sensing, computation, communication, and actuation into layered ecosystems that span edge devices, cloud infrastructures, and distributed controllers. This architectural shift redefines the scope of control theory, demanding frameworks that ensure stability, safety, and resilience across heterogeneous and dynamically reconfigurable networks (Goudarzi et al., 2022; Yu et al., 2023).

At the modeling level, CPS and IoT environments introduce hybrid dynamics, asynchronous sampling, and time-varying communication topologies. Control-theoretic approaches, such as consensus algorithms, event-triggered control, and distributed MPC, have been adapted to operate under these constraints, ensuring coordination among large populations of agents (Zhang et al., 2019). Descriptor-type and networked-system formulations capture algebraic constraints and inter-agent couplings, enabling the design of controllers that remain consistent despite latency, packet loss, or intermittent connectivity (Maxim & Caruntu, 2021).

A key challenge in these settings is the runtime adaptation. IoT-enabled control architectures increasingly rely on edge computing to host observers and estimation pipelines close to physical processes, thereby reducing latency and mitigating bandwidth limitations (Gkonis et al., 2023). Robust and resilient observers are deployed at the edge to reconstruct hidden states from sparse measurements, whereas the cloud layer provides global optimization and learning-driven adaptation. This layered deployment allows the control loop to remain operational under adversarial conditions, ranging from stochastic disturbances to cyberattacks, while ensuring scalability across thousands of interconnected devices (Islam et al., 2025).

The integration of CPS and IoT with emerging control paradigms also enables resilient and self-healing infrastructures. Observer-based anomaly detection mechanisms and runtime verification modules are embedded within networked controllers to identify faults, mis-synchronizations, or malicious intrusions in real time (Alain et al., 2025). Adaptive and distributed controllers then reconfigure the actuation

policies to maintain global stability and service continuity, ensuring that local disruptions do not propagate into systemic failures (Severson et al., 2020).

By serving as integration platforms, CPS and IoT extend the reach of control theory beyond individual plants into interconnected multi-agent ecosystems. The fusion of network-aware modeling, resilient observer designs, and distributed optimization establishes a foundation for control systems that are stable, efficient, adaptive, and secure. This paradigm positions CPS and IoT as indispensable enablers for the deployment of next-generation control architectures that must operate reliably in dynamic, uncertain, and adversarial environments (Segovia-Ferreira et al., 2024).

6.4. Quantum and edge computing for control

The advent of quantum computing and edge architectures is reshaping the computational substrates on which control systems are designed and deployed. While conceptually distinct, these paradigms converge in their potential to overcome the limitations of classical centralized computation, providing new avenues for scalability, adaptability, and real-time responsiveness in complex control environments (Furutanpey et al., 2023).

Quantum computing offers fundamentally new algorithmic tools for control design and analysis. Variational quantum algorithms (VQA) and quantum approximate optimization algorithms (QAOA) have been explored for solving high-dimensional optimal control problems, where classical approaches suffer from combinatorial complexities. Quantum linear algebra primitives enable faster solutions to Riccati- and Lyapunov-type equations that underpin linear–quadratic control and stability analysis (Sun & Zhang, 2017). Beyond optimization, quantum filtering and quantum-inspired observer designs are being investigated for descriptor and nonlinear systems, promising accelerated estimation in high-dimensional spaces. Although practical deployment remains constrained by hardware noise and scalability, the potential for hybrid quantum–classical pipelines positions quantum computing as a transformative enabler for control under intractable complexities (Clayton et al., 2024; Miao et al., 2016).

In parallel, edge computing has emerged as a practical solution to the latency and bandwidth bottlenecks that are inherent in cloud-centric deployments. By localizing computation at or near the physical process, edge platforms enable the ultra-low-latency execution of MPCs, disturbance observers, and anomaly detection schemes (Mtowe & Kim, 2023). This proximity reduces the reliance on high-throughput communication, allowing the control loops to remain operational under stringent timing constraints and adversarial network conditions (Dai et al., 2019). Moreover, edge architectures facilitate distributed coordination by hosting lightweight predictive and adaptive controllers across multiple nodes, thereby enhancing the resilience and scalability of multi-agent systems (Zhao et al., 2025).

The intersection of the quantum and edge paradigms suggests novel hybrid architectures. Quantum-accelerated solvers may be embedded within cloud- or edge-assisted supervisory layers, offloading computationally intensive optimization while preserving real-time responsiveness at the local level. Conversely, edge-deployed controllers provide the low-latency feedback required to stabilize systems while leveraging quantum resources for global coordination and large-scale predictive tasks (Bhatia & Sood, 2024). This architecture is exemplified by resource-allocation frameworks using VQA/QAOA within cloud–edge infrastructures (Mastroianni et al., 2024) and the quantum error-correcting code (QECC) model that integrates quantum, edge, and cloud computing layers to support efficient and secure IoT operations (Akter et al., 2024).

By embedding quantum algorithms into supervisory optimization and leveraging edge computing for real-time responsiveness, next-generation control frameworks achieve a dual balance between the scalability of quantum-enhanced computation and the immediacy of edge-level actuation. This integration extends the reach of control theory into domains where both dimensionality and responsiveness present fundamental barriers, highlighting a frontier where theoretical advances in computation and rigorous control design coalesce (Mastroianni et al., 2023).

6.5. Sustainability and resilience-driven control

The increasing societal emphasis on sustainability and resilience has redefined the objectives of modern control designs. Beyond the classical notions of stability and performance, control systems are now required to minimize environmental impact, optimize resource utilization, and withstand disruptions arising from faults, cyberattacks, and extreme external events (Heikal et al., 2025). These dual imperatives necessitate the reorientation of control-theoretic frameworks towards multi-objective formulations that explicitly incorporate energy efficiency, robustness, and long-term system adaptability (Aasa et al., 2025).

Sustainability in control manifests most prominently through the integration of renewable and distributed energy resources, where variability in generation demands requires controllers that can balance stability with a minimal reliance on fossil fuel-based reserves. MPC and stochastic optimization frameworks have been adapted to include carbon cost terms, energy storage constraints, and demand–response policies, thereby embedding sustainability criteria directly into the control loop (Lim et al., 2025; Yao, Liu, Chang, & Yang, 2023). In industrial manufacturing and robotics, life-cycle-aware control strategies optimize not only short-term throughput but also long-term energy and material consumption, thereby aligning closed-loop operations with broader environmental goals (Battarra et al., 2025).

Resilience extends these concepts by ensuring that control systems maintain safe operations under adverse and uncertain conditions. Observer-based anomaly detection has emerged as a critical mechanism that enables the rapid identification of faults or malicious perturbations (Xin & Long, 2025). Lyapunov-based resilient control and adaptive architectures further allow systems to recover from component failures or degraded sensor measurements while preserving their stability (Oyama & Durand, 2020). Distributed consensus protocols reinforce the resilience of multi-agent systems, allowing collective decision-making and control actions even under partial communication losses or heterogeneous disturbances (Wang et al., 2023).

The integration of sustainability and resilience objectives produces inherently multilayered control architectures. At the lower level, real-time controllers enforce operational safety and efficiency, whereas supervisory layers optimize long-term resource allocation under environmental and resilience constraints (Rehmat et al., 2025). Emerging paradigms, including edge-enabled predictive controllers and hybrid physics–data models, offer the computational and modeling flexibility required to implement such multi-objective strategies in practice (Arroba et al., 2024; Wu et al., 2019).

By embedding sustainability and resilience as first-class objectives, control theory extends its role from guaranteeing system stability to shaping environmentally conscious, fault-tolerant, and future-proof infrastructures. This evolution reflects changing societal priorities and highlights the adaptability of the control-theoretic toolbox to address challenges at the intersection of engineering, environment, and security.

Table 9 consolidates the integration of emerging paradigms with control theory, highlighting how each technological strand reshapes the design and implementation of feedback architecture. The table compares paradigms ranging from data-driven learning and digital twin synchronization to distributed IoT/CPS frameworks, quantum-inspired formulations, and sustainability-driven optimizations. Across all cases, observer-based estimation remains a common denominator, enabling resilient monitoring under uncertainty and bridging the physical dynamics with computational models. The readiness column indicates the maturity of each paradigm, from exploratory research in quantum

control to pilot deployments of digital twins and operational IoT infrastructures. Representative domains underscore the sectoral diversity of these integrations, from energy and aerospace to healthcare and robotics, reflecting the breadth and heterogeneity of current adoption trajectories.

The comparative analysis underscores that while control theory has successfully expanded into domains enriched by ML, cyber–physical integration, and advanced computation, the path towards widespread adoption remains contingent on overcoming significant challenges. Issues of scalability, verifiability, interpretability, and resilience under adversarial or uncertain conditions remain unresolved, particularly as systems become more interconnected and autonomous. Furthermore, questions regarding standardization, certification, and sustainable deployment highlight the need for rigorous frameworks that can reconcile theoretical guarantees with practical constraints. These open fronts form the foundation for the following section, which critically examines the challenges and future trends that will shape the next era of control theory.

7. Challenges and future trends

As technology and systems become increasingly complex, control theory is evolving to meet the new demands and capitalize on emerging opportunities. Future trends point towards the integration of advanced technologies, interdisciplinary applications, and the development of adaptive and intelligent control systems. However, these advancements present challenges that require innovative solutions, particularly in ensuring the reliability, resilience, and ethical deployment of these technologies. This section explores the emerging trends that are shaping the future of control theory and highlights the challenges that must be addressed to unlock its full potential.

7.1. Challenges

Despite its potential, control theory faces significant challenges in evolving to meet the demands of modern systems. One of the foremost challenges is managing the increasing complexity of systems that integrate diverse components and operate at multiple scales. For example, in large-scale systems such as smart grids or autonomous transportation networks, controllers must process vast amounts of data, maintain real-time performance, and ensure scalability without compromising reliability (Benner, 2004; De Bot, 2017; Majumdar et al., 2020).

A significant difficulty arises in nonlinear observer design, particularly when global state estimation is required under strongly nonlinear and uncertain conditions. Laguerre polynomials provide a powerful approximation framework; however, their use introduces computational burdens in high-dimensional systems and complicates the tuning of observer gains (González-Prieto & Villaverde, 2021; Lee & Chang, 1986; Wang, 2004). Although advances in adaptive gain optimization and high-performance computing have alleviated some limitations, achieving real-time performance without sacrificing robustness remains an open research frontier (Boulghasoul et al., 2014; Gheorghiu & Gheorghiu, 2014; Perng, 1986).

Resilience and adaptability are critical concerns. Systems must be designed to withstand disruptions caused by environmental factors, cyberattacks, or component failures. This requires robust algorithms capable of anomaly detection, fast recovery, and autonomous adaptation while operating under the computational constraints of embedded or edge platforms (Baeten et al., 2011; Das et al., 2020; Martin et al., 2015; Yodo et al., 2017).

Another recurring challenge is the noise sensitivity of the widely deployed PID controllers. In particular, the derivative term ($K_D \frac{de(t)}{dt}$) amplifies the high-frequency noise, destabilizing the loop and degrading the performance. To counteract this effect, low-pass filters are incorporated into the derivative channel; however, achieving the right balance between disturbance attenuation and control responsiveness is nontrivial (Bies et al., 2023; Chen et al., 2018; Segovia et al., 2014). This highlights the broader difficulty in translating theoretically simple control laws into robust industrial implementations.

Ethical and governance issues further complicate the landscape, especially as AI-enabled controls become pervasive. Beyond the general principles of transparency and fairness, formal standards now provide structured guidance, such as IEEE 7000-2021, the EU AI Act, and ISO/IEC 23894:2023 on AI risk management (Agbese et al., 2023; Laux et al., 2024; Oviedo et al., 2024). These frameworks emphasize risk assessment, human oversight, and bias mitigation; however, their integration into real-time safety-critical control loops remains immature. In high-risk domains such as autonomous driving and healthcare robotics, explainability, liability attribution, and patient consent require not only compliance with ISO 26262 and FDA SaMD guidelines but also continuous auditing and dynamic ethical risk modeling (Iyenghar et al., 2024; Lal et al., 2022; Porter et al., 2024).

Ultimately, bridging the gap between theoretical sophistication and practical deployment remains a persistent challenge. Hardware limitations, lack of interoperability standards, and insufficient interdisciplinary collaboration often prevent advanced algorithms from being adopted in industrial or clinical settings (Ericsson et al., 2016; Pickering, 2023; Rantzer, 2015).

Table 10 synthesizes the principal categories of challenges that modern control theory must confront, ranging from computational and modeling barriers to resilience, ethical accountability, and translational gaps to clinical practice. The table highlights how technical obstacles, such as the computational burden of nonlinear observers and noise sensitivity of PID controllers, coexist with broader system-level demands for scalability, adaptability, and trustworthy operation. By mapping each challenge to its defining concerns and representative references, the table provides a consolidated perspective that complements the narrative discussion and clarifies the multidimensional hurdles that shape current and future research trajectories.

Addressing these issues requires tighter collaboration between theorists, engineers, and policymakers to ensure that advances in control theory can migrate smoothly from simulations to real-world systems. The need to align theoretical innovation with deployment readiness directly motivates the future trends discussed in the following subsection.

7.2. Future trends

The future of control theory is marked by its convergence with cutting-edge technologies and applications in emerging fields. One of the most significant trends is the integration of AI and ML into control systems. These technologies enable controllers to learn and adapt to new environments and conditions without explicit programming, making them particularly effective for systems that operate in dynamic and uncertain conditions. RL is gaining traction for its ability to optimize control strategies through trial and error, providing solutions for highly complex and nonlinear systems (Bensoussan et al., 2022; Bertsekas, 2019; Weyns et al., 2021).

Another critical trend is the rise of CPS and IoT, which are driving the need for distributed control mechanisms. Unlike current deployments, which mostly focus on interoperability, future research is expected to emphasize adaptive, self-organizing controllers capable of managing heterogeneous infrastructures in real time. For instance, in smart cities, control systems coordinate transportation, energy, and water management infrastructure to enhance efficiency and sustainability (Bordel et al., 2017; Crisostomi et al., 2016; Radisavljevic-Gajic et al., 2017; Sanfelice et al., 2016; Shastri et al., 2008).

Quantum control is an emerging, transformative field that addresses the unique challenges of quantum systems, including the preservation of coherence, error mitigation, and precise state manipulation. In the short term, realistic applications focus on quantum error correction protocols, quantum sensor calibration, and quantum gate optimization,

Table 9
Comparative summary of emerging paradigms integrated with control theory.

Emerging paradigm	Role in control	Observer/Controller integration	Deployment readiness	Representative domains
AI and ML (Aghajari & AbdulRahim, 2024; Bhatti et al., 2025; Cao et al., 2024; Dritsas & Trigka, 2025a, 2025b; Greatwood & Richards, 2019; Joglekar et al., 2022; Khan, Cao, Li, & Li, 2021; Lawrence et al., 2024; Peng et al., 2022; Pillonetto et al., 2025; Stiti et al., 2024; Weber et al., 2024; Wu et al., 2025; Yu et al., 2025; Zhu et al., 2023)	Reinforcement, supervised and bio-inspired learning for adaptive and predictive control; deep and neural dynamic networks for nonlinear approximation, real-time constrained optimization, and predictive fault detection	Neural observers, physics-informed learning hybrids, and BAS–ZNN-inspired solvers for adaptive and predictive control integration	Early industrial adoption in automotive and robotics; simulation and hybrid test validation for embedded CPS and intelligent energy systems	Transportation, robotics, process control, energy systems, financial optimization, adaptive control frameworks
Digital Twins and Data-Driven Modeling (Bárkányi et al., 2021; Gratius et al., 2023; Kapteyn et al., 2022, 2021; Lei et al., 2023; Mchirgui et al., 2024; Peterson et al., 2025; Reich, 2025; Van Dinter et al., 2022; Yao, Yang, Wang, & Zhang, 2023)	Real-time synchronization of physical and virtual plants; predictive what-if analysis for fault anticipation	Coupling with MPC and LQR; observer-based synchronization of twin-plant states	Industrial pilots in energy, aerospace, and manufacturing; maturing standards for interoperability	Energy grids, aerospace, manufacturing
CPS and IoT (Alain et al., 2025; Gkonis et al., 2023; Goudarzi et al., 2022; Islam et al., 2025; Maxim & Caruntu, 2021; Segovia-Ferreira et al., 2024; Severson et al., 2020; Yu et al., 2023; Zhang et al., 2019)	Networked coordination with latency and uncertainty; distributed resilience and anomaly detection	Resilient observers for packet drops and attacks; consensus controllers for multi-agent systems	Widely deployed in industrial IoT; emerging in safety-critical infrastructures	Industrial automation, smart cities, distributed energy
Quantum and Edge Computing (Akter et al., 2024; Bhatia & Sood, 2024; Clayton et al., 2024; Dai et al., 2019; Furutanpey et al., 2023; Mastroianni et al., 2024, 2023; Miao et al., 2016; Mtowe & Kim, 2023; Sun & Zhang, 2017; Zhao et al., 2025)	Exploiting quantum algorithms for predictive control and optimization; edge devices for low-latency execution	Variational quantum controllers; edge-deployed observers for real-time estimation	Mostly exploratory in quantum; early embedded implementations at the edge	Smart grids, autonomous systems, secure communications
Sustainability and Resilience-Driven Control (Aasa et al., 2025; Arroba et al., 2024; Battarra et al., 2025; Heikal et al., 2025; Lim et al., 2025; Oyama & Durand, 2020; Rehmat et al., 2025; Wang et al., 2023; Wu et al., 2019; Xin & Long, 2025; Yao, Liu, Chang, & Yang, 2023)	Multi-objective optimization balancing stability, energy efficiency, and environmental impact; resilience under faults and attacks	Observer-based anomaly detection; Lyapunov-resilient controllers; distributed consensus for recovery	Early integration in energy and manufacturing; policy-driven expansion in infrastructure	Power systems, healthcare devices, industrial robotics

where control techniques enhance the robustness against noise and operational imperfections. Compared to classical methods, quantum control operates in probabilistic, high-dimensional spaces, requiring fundamentally different strategies based on unitary transformations and quantum-level feedback. Although still in its infancy, this field already shows disruptive potential by demonstrating measurable advantages in quantum sensing, secure communication, and small-scale processors. Ding et al. (2018), Kallush et al. (2022), Koch et al. (2022), Mansell (2020).

One promising direction in nonlinear observer design is the utilization of Laguerre polynomials for state estimation in such systems. This approach addresses the limitations of traditional methods, such as local linearization and HGO, by offering a globally valid approximation. Laguerre-based techniques enable robust state estimation in systems characterized by strong nonlinearities and dynamic uncertainties, making them particularly suitable for applications in robotics, autonomous vehicles, and smart grids. The orthogonal basis of the Laguerre polynomials simplifies the design and analysis of observers, facilitating their implementation in high-dimensional and real-time environments. Although the current work identifies computational challenges (as noted in the previous subsection), future developments may reframe Laguerre-based observers as a scalable paradigm for resilient and adaptive state estimation. De Figueiredo and Dwyer (1980), Keighobadi

Table 10
Key challenges of control theory in modern systems.

Category	Key points	References
Managing Complexity	- Handling large-scale, heterogeneous systems with vast data streams - Ensuring scalability and real-time performance without compromising reliability	De Bot (2017) Majumdar et al. (2020) Benner (2004)
Laguerre-based Nonlinear Observer Design	- Global state estimation under nonlinear, uncertain dynamics - Computational burden in high-dimensional systems - Need for adaptive gain optimization and high-performance computation - Real-time robustness remains an open frontier	Lee and Chang (1986) González-Prieto and Villaverde (2021) Wang (2004) Boulghasoul et al. (2014) Gheorghiu and Gheorghiu (2014) Perng (1986)
Resilience & Adaptability	- Designing controllers to detect anomalies, recover rapidly, and adapt autonomously - Challenge of balancing resilience with computational constraints in embedded/edge platforms	Yodo et al. (2017) Martin et al. (2015) Das et al. (2020) Baeten et al. (2011)
Noise Management in PID Controllers	- Derivative term amplifies high-frequency noise - Low-pass filters mitigate disturbances but may reduce responsiveness - Balancing robustness and agility remains nontrivial	Segovia et al. (2014) Bies et al. (2023) Chen et al. (2018)
Ethical & Governance Considerations	- Embedding transparency, fairness, and privacy in AI-enabled control - Application of standards (IEEE 7000-2021, EU AI Act, ISO/IEC 23894:2023) - Sector-specific needs in autonomous driving and healthcare: explainability, liability, consent - Integration of continuous auditing and dynamic ethical risk modeling	Agbese et al. (2023) Laux et al. (2024) Oviedo et al. (2024) Iyenghar et al. (2024) Lal et al. (2022) Porter et al. (2024)
Bridging Theory & Practice	- Barriers from hardware limitations and lack of interoperability standards - Necessity of interdisciplinary collaboration to enable deployment readiness	Pickering (2023) Ericsson et al. (2016) Rantzer (2015)

et al. (2021), Mirinejad and Inanc (2017), O'Leary et al. (2015), Zhang et al. (2024), Zhong et al. (2018).

Finally, digital twin technology is advancing towards fully autonomous and adaptive decision-making by integrating AI and ML models within virtual representations of physical assets. These digital twins are becoming increasingly dynamic and capable of predictive diagnostics, real-time optimization, and resilient control in domains such as autonomous vehicles, aerospace systems, and smart energy grids. The integration of the IoT enables the generation of high-fidelity, continuously updated digital twins. A realistic near-term roadmap envisions digital twins evolving towards greater interoperability, modularity, and security. In particular, future convergence with quantum and edge computing is expected to enhance the scalability and computational responsiveness of digital twin systems, enabling real-time adaptive control in complex, distributed environments. In this vision, digital twins will serve as unifying platforms where advanced computation, system resilience, and trustworthy autonomy converge, providing a cornerstone for next-generation intelligent control (Acharya et al., 2024; Heluany & Gkioulos, 2023; Kharchenko et al., 2020; Mostafa et al., 2021; Paul et al., 2024).

Table 11 consolidates the major future trajectories of control theory by linking emerging computational paradigms to domain-specific methodological advances. This comparison highlights how ML integration fosters adaptivity in nonlinear and uncertain environments, whereas CPS and IoT drive decentralization and cross-domain coordination. Quantum control stands out as a paradigm shift, requiring fundamentally different strategies than classical methods, and has already demonstrated tangible advantages in precision sensing and secure communication. Meanwhile, the Laguerre polynomial-based observer design illustrates how mathematically rigorous tools can extend scalability and robustness in real-time estimation. Finally, digital twin technologies epitomize the convergence of cyber and physical layers, evolving towards autonomous, AI-driven, and resilient control ecosystems supported by edge and quantum computing. Collectively, the table underscores that the frontier of control theory will be shaped less by isolated techniques and more by their integration with disruptive technologies.

The trends outlined above emphasize that the future of control theory lies in its capacity to transcend disciplinary boundaries and integrate seamlessly with AI, quantum computation, and cyber–physical infrastructure. What unites these diverse directions is the pursuit of resilience, adaptability, and ethical accountability in increasingly uncertain environments that are constantly changing. As these trajectories converge, they redefine the role of control theory, not merely as a mathematical discipline, but as a foundational enabler of intelligent, networked, and sustainable systems. This forward-looking perspective naturally transitions into the concluding section, where the implications of these developments are synthesized and framed within the broader landscape of theory, practice, and societal impact.

8. Discussion

This survey extends the scope of previous studies in control theory by presenting a unified cross-domain framework that links foundational methods with emerging intelligent paradigms, as shown in Table 12. Earlier surveys have typically focused on narrow slices of the field, such as quantum control (Dong & Petersen, 2010), the transition from model-based to data-driven paradigms (Hou & Wang, 2013), educational experimentation (Horáček, 2000), single benchmarks such as the inverted pendulum (Boubaker, 2013), or specialized areas such as repetitive control (Hillerström & Walgama, 1996), operational applications in supply chains and Industry 4.0 (Ivanov et al., 2018), and comprehensive review analysis of N-PID control (Çelik et al., 2026) and FOPID control (Bashishtha et al., 2025) advancements and applications. In contrast, this survey integrates historical fundamentals, including dynamic systems modeling and stability analysis, control design principles (e.g., PID, LQR/LQG, MPC, robust), and modern developments such as AI/ML-enhanced control, CPS/IoT architectures, quantum control frameworks, and digital twins. A distinctive contribution is the mathematically rigorous treatment of nonlinear observer design using

Table 11
Future trends towards the integration of advanced technologies for the development of intelligent control systems.

Category	Key points	References
AI & ML Integration	- Enable controllers to adapt without explicit programming - Employ RL for complex, nonlinear environments - Advanced data-driven observers and hybrid physics–AI control architectures	Bensoussan et al. (2022) Weyns et al. (2021) Bertsekas (2019)
CPS & IoT	- Develop distributed control for heterogeneous infrastructures - Enable adaptive, self-organizing coordination across domains - Drive smart city ecosystems in energy, transport, and water management	Bordel et al. (2017) Sanfelice et al. (2016) Radisavljevic-Gajic et al. (2017) Crisostomi et al. (2016) Shastri et al. (2008)
Quantum Control	- Advance error correction, quantum sensor calibration, and gate optimization - Operate in probabilistic, high-dimensional spaces distinct from classical methods - Demonstrate early advantages in sensing, secure communication, and small-scale processors	Kallush et al. (2022) Mansell (2020) Ding et al. (2018) Koch et al. (2022)
Nonlinear Observer Design with Laguerre Polynomials	- Provide globally valid approximations for nonlinear dynamics - Simplify observer analysis through orthogonal polynomial bases - Enhance robustness against uncertainties and noise in real time - Extend scalability to robotics, autonomous systems, and smart grids	De Figueiredo and Dwyer (1980) Keighobadi et al. (2021) Zhang et al. (2024) Mirinejad and Inanc (2017) Zhong et al. (2018) O'Leary et al. (2015)
Digital Twins	- Evolve towards autonomous, AI-driven platforms for diagnostics and control - Integrate with edge and quantum computing to boost scalability and responsiveness - Address interoperability, secure synchronization, and lightweight deployment challenges	Mostafa et al. (2021) Kharchenko et al. (2020) Paul et al. (2024) Acharya et al. (2024) Heluany and Gkioulos (2023)

Laguerre polynomials, supported by a direct comparative analysis with the EKF, HGO, and hybrid stochastic estimation approaches. This combination of breadth and depth uniquely positions the survey to serve as both a consolidated reference and a forward-looking guide for research.

From a control strategy perspective, the analysis confirms the persistent relevance of PID and LQR controllers in structured, predictable environments, while demonstrating the necessity of MPC, fault-tolerant control, and RL in dynamic, uncertain, and multi-agent contexts. This shift represents a paradigmatic transition from controllers as static stabilizers to controllers as adaptive, learning-based agents capable of anticipating and optimizing performance under variable conditions.

The implications for practice are domain-specific and technically applicable. In industrial automation, adaptive control mechanisms combined with predictive fault tolerance enable high-precision and resilient manufacturing processes. In transportation systems, polynomial-based nonlinear observers support accurate vehicle state estimation under strong nonlinearities, thereby directly enhancing the safety of autonomous navigation. In healthcare, adaptive state estimation integrated into medical devices improves robustness in the presence of patient-specific physiological variations. In energy and environmental systems, predictive and distributed control frameworks improve resource allocation and stability in the presence of stochastic disturbance. Across these domains, the common thread is that control is no longer only about stability but also about resilience and predictive adaptation under uncertainty. This resonates with the theoretical contribution of Laguerre-based nonlinear observers, which provide a globally valid and computationally efficient foundation for such resilience-oriented architectures.

This survey deliberately focuses on practice-oriented developments that are directly linked to deployment, rather than purely abstract theoretical advances. While this perspective emphasizes applicability and integration, it inevitably leaves certain theoretical dimensions outside the scope of review. Moreover, the rapid pace of AI-control convergence suggests that some technological details will require periodic updates to remain state-of-the-art.

At a technical level, several open challenges remain. Achieving scalability and computational efficiency continues to be difficult for large-scale or resource-constrained systems, where distributed architectures and model reduction are often necessary trade-offs. The generalization of data-driven and hybrid controllers beyond the conditions of their training data also poses inherent limitations, particularly under uncertainty, nonstationarity, or domain shifts. Addressing these issues will require future research that balances tractability, robustness, and adaptability across heterogeneous operational contexts.

In summary, this survey highlights a dual trajectory: the consolidation of robust classical control principles and the rise of intelligent, adaptive, and ethically aware architectures. Looking ahead, the defining features of next-generation control systems will be scalability, real-time adaptability, cross-domain interoperability, and governance-aware design, which can only be achieved through unified hybrid architectures that blend classical robustness with emerging adaptive intelligence.

9. Conclusions

Control theory has evolved from its classical roots in feedback, stability analysis, and state-space representation into a discipline that now operates at the forefront of intelligent, networked, and ethically aware systems. This survey presents a unified cross-domain perspective that consolidates robust classical principles with advanced methodologies, such as nonlinear observer design using Laguerre polynomials, while extending the integration with AI/ML, CPS/IoT, digital twins, and quantum-enhanced computation. By bridging historical depth with forward-looking breadth, this study demonstrates how control theory can continue to serve as a foundational pillar for engineering systems in the 21st century.

This overview underscores that while PID and LQR remain indispensable for structured environments, the future lies in architectures that are adaptive, distributed, and data-driven, and capable of operating under uncertainty, nonlinearity, and adversarial disruptions. Emerging techniques, such as RL controllers, neural observers, and hybrid physics–AI approaches, point towards the convergence of rigorous mathematics and empirical intelligence, creating systems that are not only stable but also self-optimizing and resilient.

From an application standpoint, modern control systems underpin industrial automation, autonomous transportation, healthcare technologies, and resilient energy infrastructures. Concrete examples include predictive maintenance in manufacturing, resilient power management in renewable grids, and adaptive control of medical and assistive robotics. These scenarios demonstrate how classical stability concepts are now being operationalized through learning-based estimation and distributed coordination frameworks. These domains collectively define the frontier where modeling fidelity must coexist with computational efficiency and resilience to cyber–physical uncertainty.

Table 12
Side-by-side comparison of scope, coverage, and distinctive aspects across related surveys.

Survey	Scope	Techniques covered	Applications covered	Advanced nonlinear observer coverage	Integration with emerging technologies	Distinctive features/open aspects
Çelik et al. (2026)	Advancements in N-PID controllers	✓	✓	✗	✗	In-depth review of PID control, advantages and applications./Explore N-PID in new emerging fields (e.g., renewable energy and autonomous systems).
Dong and Petersen (2010)	Quantum control theory for open and closed systems and their physical realizations	✓	✗	✗	✗	Strong focus on quantum physics; limited scalability to many-body systems and sensitive to noise
Hou and Wang (2013)	Transition from model-based to data-driven control paradigms	✓	✓	✗	✓	Early view of data-driven control; lacks integration with physics-based priors and robust stability guarantees
Horáček (2000)	Laboratory experimentation for control education	✓	✗	✗	✗	Pioneer in hands-on labs; no CPS/IoT or AI-enabled labs, limited scalability for remote education
Boubaker (2013)	Inverted pendulum as a benchmark for nonlinear control	✓	✓	✓	✗	Widely used benchmark; limited transferability beyond pendulum, sensitive to noise
Hillerström and Walgama (1996)	Repetitive control theory for periodic reference tracking and disturbance rejection	✓	✓	✗	✗	Focused on periodic control; scalability to MIMO systems and quasi-periodic tasks remains open
Ivanov et al. (2018)	Control in operational systems, supply chains, and Industry 4.0	✓	✓	✗	✓	Industry 4.0 orientation; open issues in real-time data integration and cybersecurity
Bashishtha et al. (2025)	Detailed analysis in FOPID controllers	✓	✓	✗	✗	Wide FOPID applications, classifications, and objective functions
This survey	Unified cross-domain coverage of classical control, advanced nonlinear observers, and integration with AI, CPS, IoT, quantum, and digital twins	✓	✓	✓	✓	Combines breadth (cross-domain) with depth (Laguerre-based nonlinear observers) and future-facing tech integration; excludes purely theoretical advances, requires periodic updates due to fast AI-control evolution

Looking ahead, key research directions include the development of unified benchmarks for evaluating hybrid control–AI architectures, scalable observer–learning frameworks that preserve interpretability, and the systematic integration of real-time optimization with safety-critical guarantees. Progress in these directions will accelerate the transition from laboratory prototypes to reliable, field-deployable intelligent control systems. Future innovation will depend on the tighter integration of control with high-performance computing, edge/quantum platforms, and real-time data ecosystems, ensuring that system-level intelligence remains scalable, interoperable, and secure.

Beyond technical progress, control theory must engage with the broader imperatives of sustainability, governance, and ethical responsibility. The design of future control systems must incorporate the principles of fairness, transparency, accountability, and environmental stewardship, ensuring that technological advances align with societal goals.

In conclusion, the trajectory outlined in this survey reveals a dual imperative: to preserve and extend the mathematical rigor of classical control while embracing the adaptability and intelligence of emerging paradigms. Meeting this challenge requires collaborative efforts from theorists, engineers, computer scientists, and policymakers. If realized, the next generation of control systems will not only advance science and technology but also serve as a cornerstone for resilient, sustainable, and human-centered infrastructure worldwide.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Dr Maria Trigka received her Diploma degree from the Department of Electrical and Computer Engineering and her Master and Ph.D. degrees in Signal Processing for Wireless Communications from the Department of Computer Engineering and Informatics, University of Patras, Greece. Her general research interests span the scientific areas of statistical signal processing and learning. She is the author and co-author of publications in the areas of Signal Processing, Wireless Communications, and Machine Learning. As a PhD candidate, she served as a research scholar under the project "Strengthening Human Research Potential through Doctoral Research" co-financed by the Greece — State Scholarships Foundation (IKY) and the European Union. She has been involved in numerous research projects co-financed by the European Union and Greek national funds.



Dr. Elias Dritsas received his Diploma, M.Sc., and PhD degrees in Computer Science and Informatics from the Department of Computer Engineering and Informatics, University of Patras, Greece. He also earned an MBA from the University of Derby in the United Kingdom. His research interests include AI and ML, big data, databases, data mining, cloud computing, and human–computer interaction. He is the author and co-author of publications in the areas of ML and Data Analysis. He has participated in numerous development projects funded by the European Commission (EC) and the Greek Secretariat for Research and Technology. He serves as a regular reviewer for several technical journals and at conferences. He is a member of the Technical Chamber of Greece and the Association for Computing Machinery (ACM).